

**MBBS IV Junior Clerkship (2023/24)**  
**Whole Class Session 36**

# **Back Pain in an Elderly Woman: Osteoporosis and Related Fractures**

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Queen Mary Hospital

# Case scenario

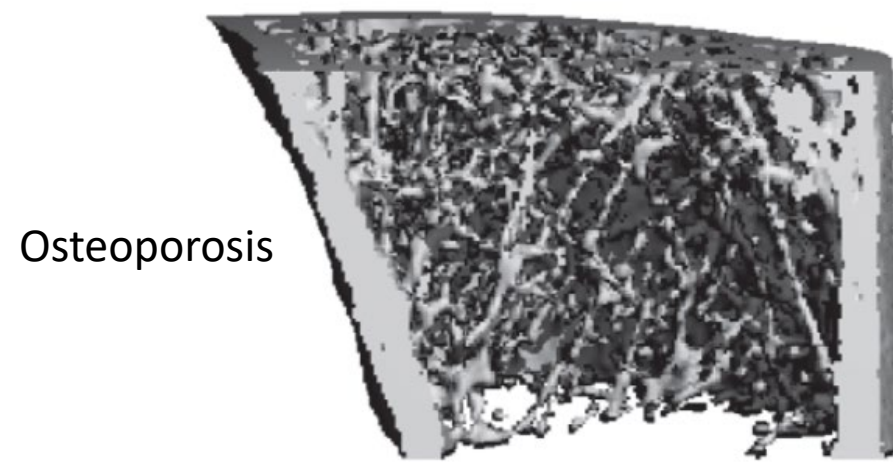
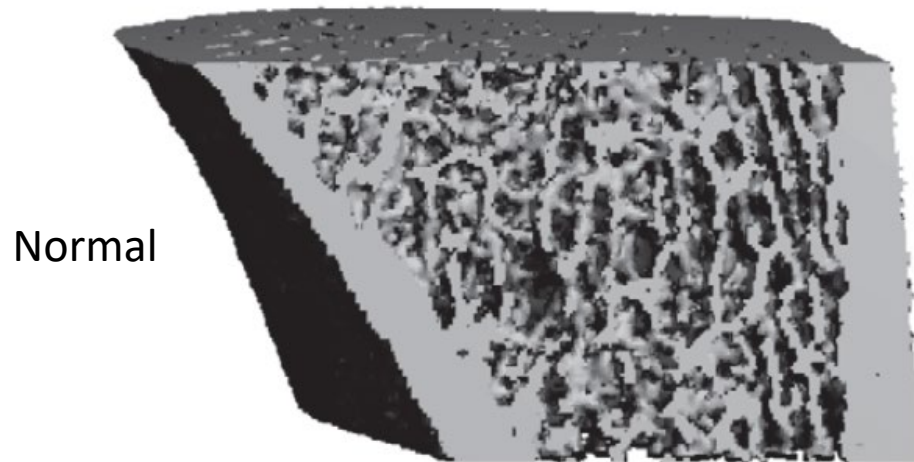
- A 70-year-old woman complained of low back pain for the recent few days
- Question: does she have osteoporosis?

# Learning objectives

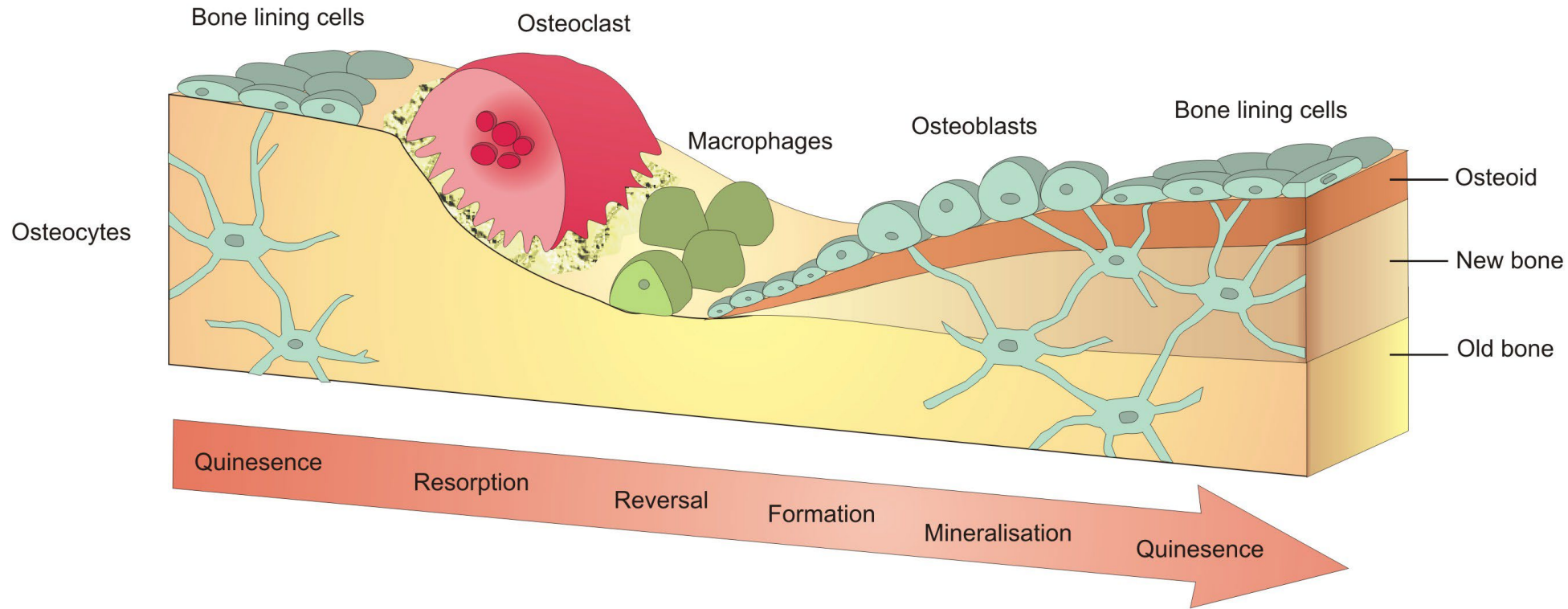
- Understand the common types of osteoporotic fractures and their clinical presentation
- Learn the epidemiology of osteoporotic fracture
- Understand the diagnosis of osteoporosis and measurement of bone mass
- Recognize the risk factors for osteoporosis
- Understand the prevention and treatment of osteoporosis
- Learn the management of osteoporotic fractures

# What is osteoporosis?

- A systemic skeletal disease characterised by **low bone mass** and **microarchitectural deterioration** of bone tissue, with a consequent increase in **bone fragility** and susceptibility to **fracture**



# Bone remodelling

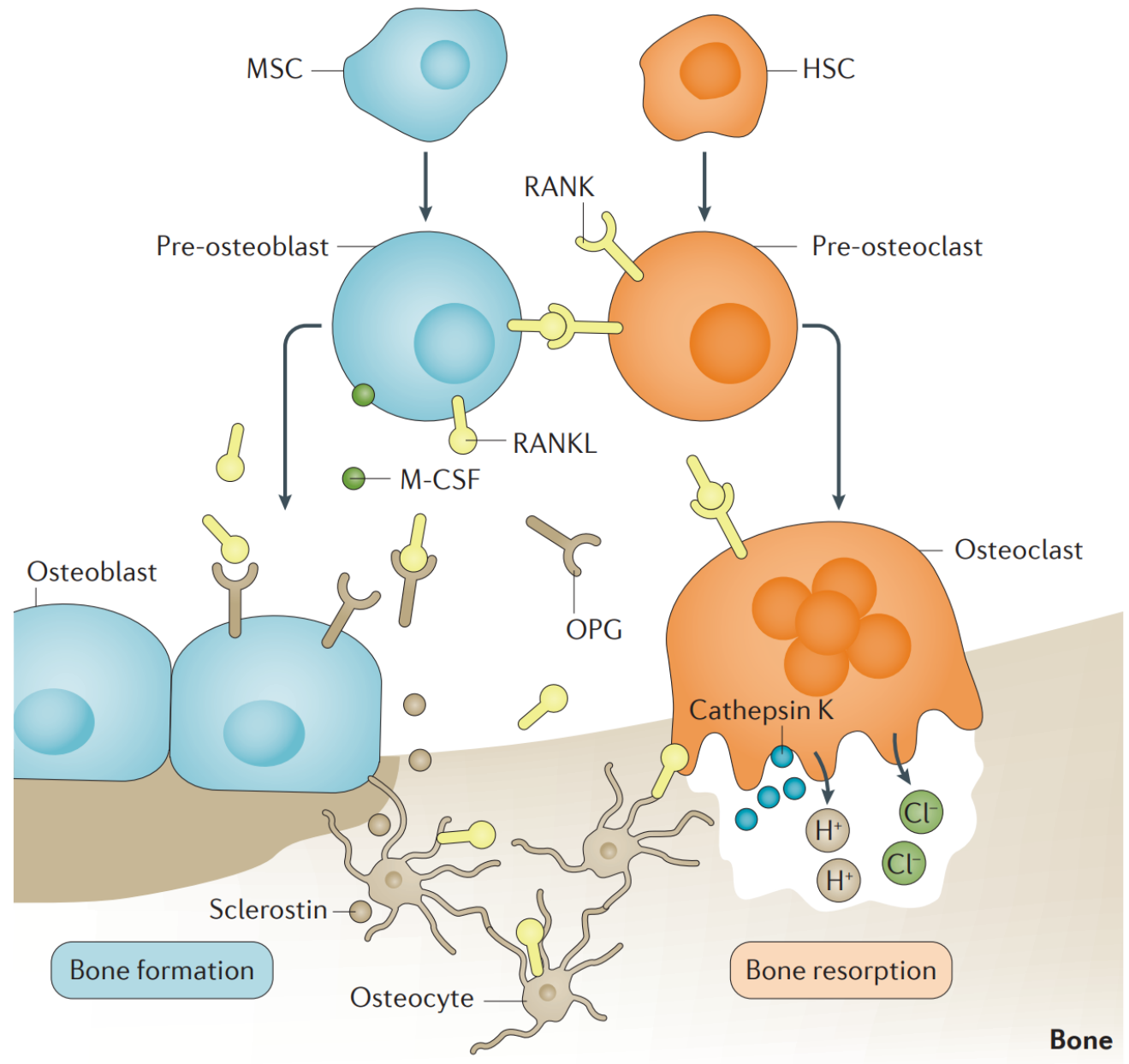


## The bone remodelling process.

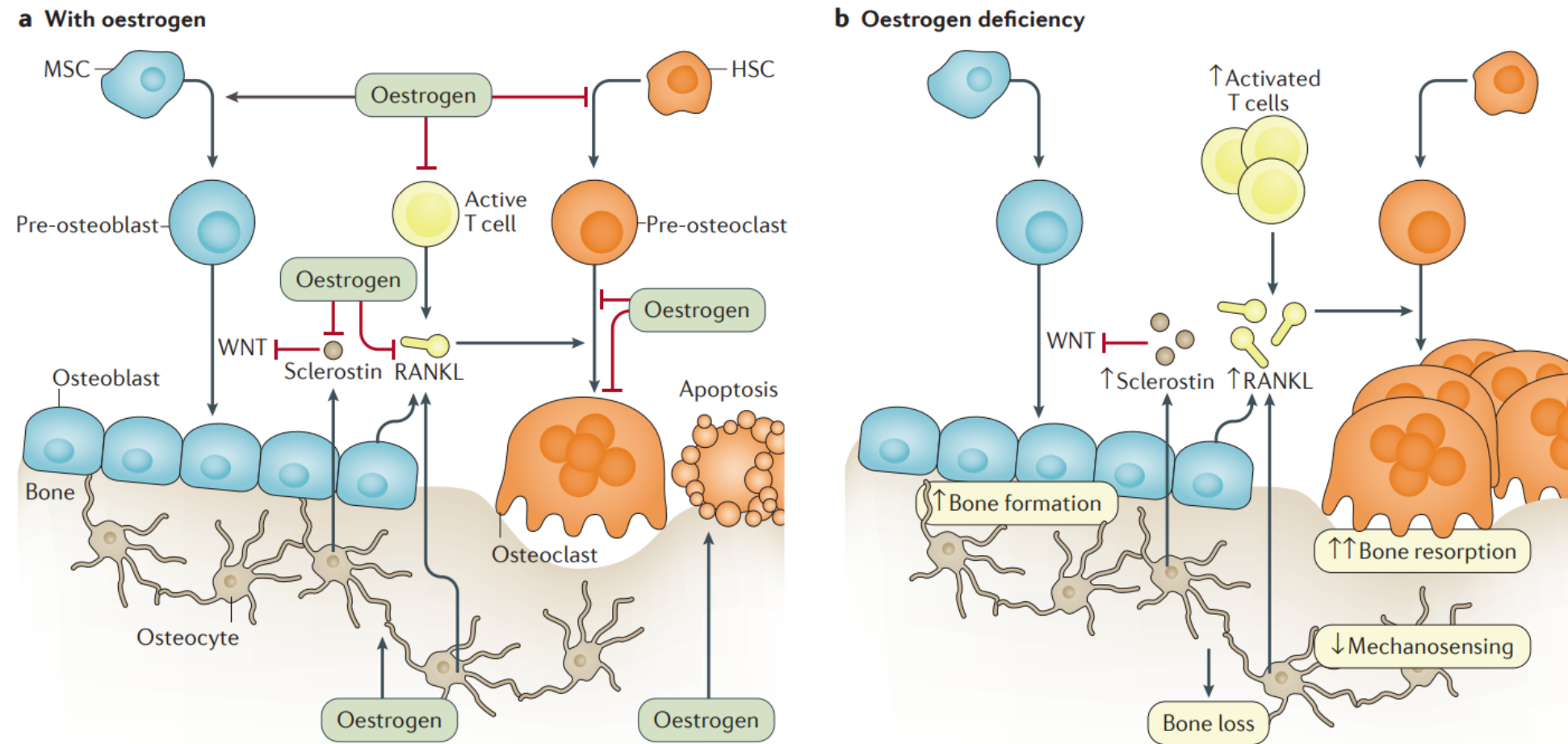
Bone is continuously remodelled at discrete sites in the skeleton in order to maintain the integrity of the tissue. During this process, old bone is resorbed by osteoclasts and replaced with new osteoid, secreted by osteoblasts. First osteoclasts are activated, and the resorption phase takes approximately 10 days. Following resorption, unclassified macrophage-like cells are found at the remodelling site in the intermediate, or reversal phase. Osteoblast precursors are then recruited, which proliferate and differentiate into mature osteoblasts, before secreting new bone matrix. The matrix then mineralises to generate new bone and this completes the remodelling process. Copyright BTR©

# Bone remodelling is coupled

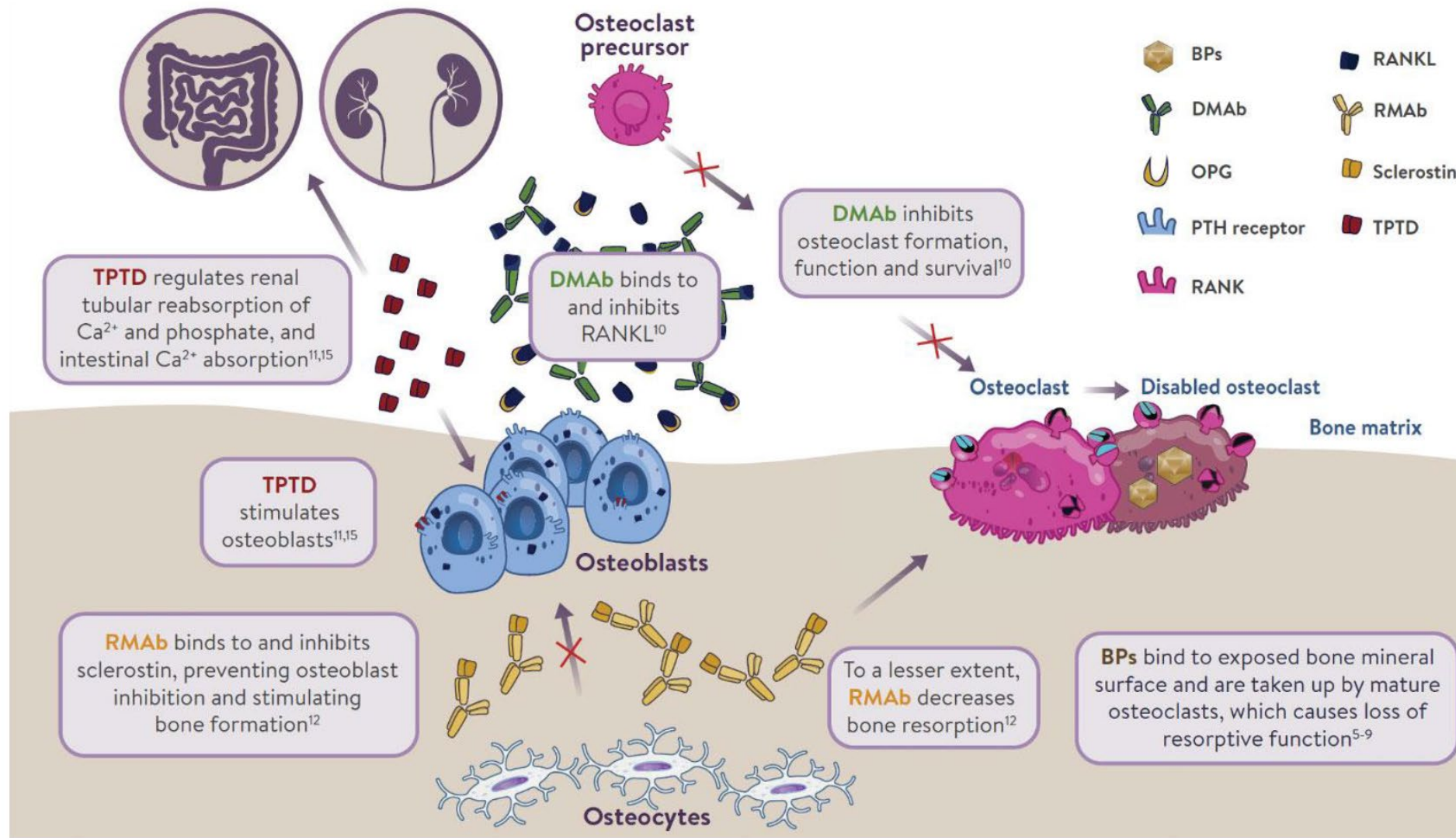
- In the adult skeleton, bone is constantly remodelled at multiple sites by the coordinated and sequential actions of bone-resorbing osteoclasts and bone-forming osteoblasts to repair microdamage and to adapt bone structure to the mechanical and metabolic needs
- Osteocytes represent >95% of all bone cells, are embedded in the mineral, function as mechanosensors and fine-tune bone remodelling by osteoblasts and osteoclasts







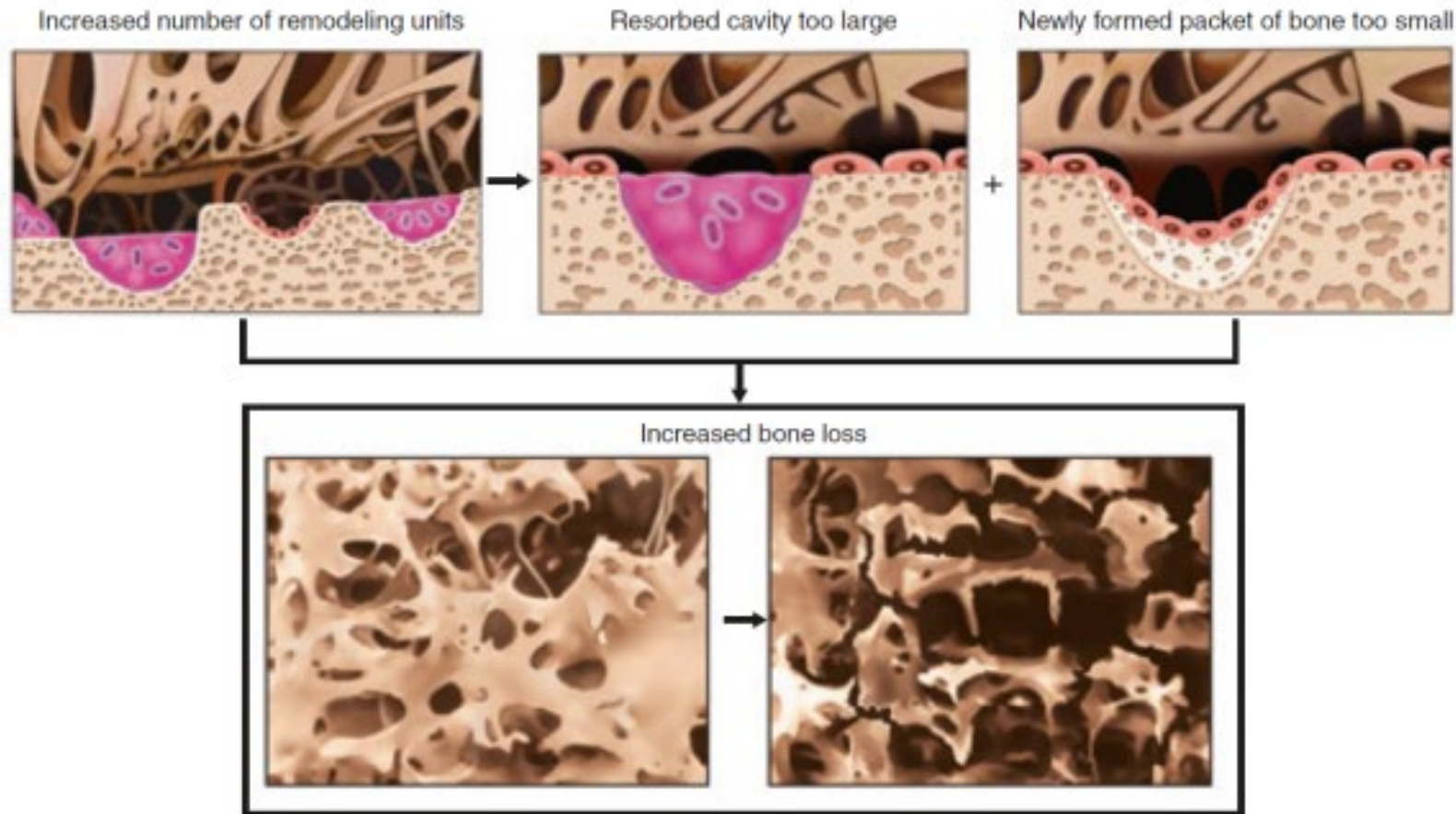
**Figure 5 | Role of oestrogens in bone remodelling. a** | Physiological bone remodelling. Oestrogens enhance bone formation by stimulating osteoblast differentiation and function and suppress osteoclast differentiation and activity through direct cellular effects and reduced production of receptor activator of nuclear factor- $\kappa$ B ligand (RANKL) by activated T cells, mesenchymal stem cells (MSCs), osteoblasts and osteocytes. In addition, oestrogens are required for adequate mechanosensing and mechanotransduction by the osteocytes. **b** | Oestrogen deficiency. RANKL production is upregulated translating into a higher osteoclast number and activity. Concurrently, the function of individual osteoblasts is suppressed, partly by upregulation of the WNT inhibitor sclerostin, whereas the total number of osteoblasts increases. The role of osteocytes in mechanosensing is also impaired. HSC, haematopoietic stem cell.



**Fig. 1.** Mechanism of action of osteoporosis therapies. BP, bisphosphonate; DMAB, denosumab; OPG, osteoprotegerin; PTH, parathyroid hormone; RANK, receptor activator of nuclear factor kappa-B; RANKL, receptor activator of nuclear factor kappa-B ligand; RMAb, romosozumab; TPTD, teriparatide.

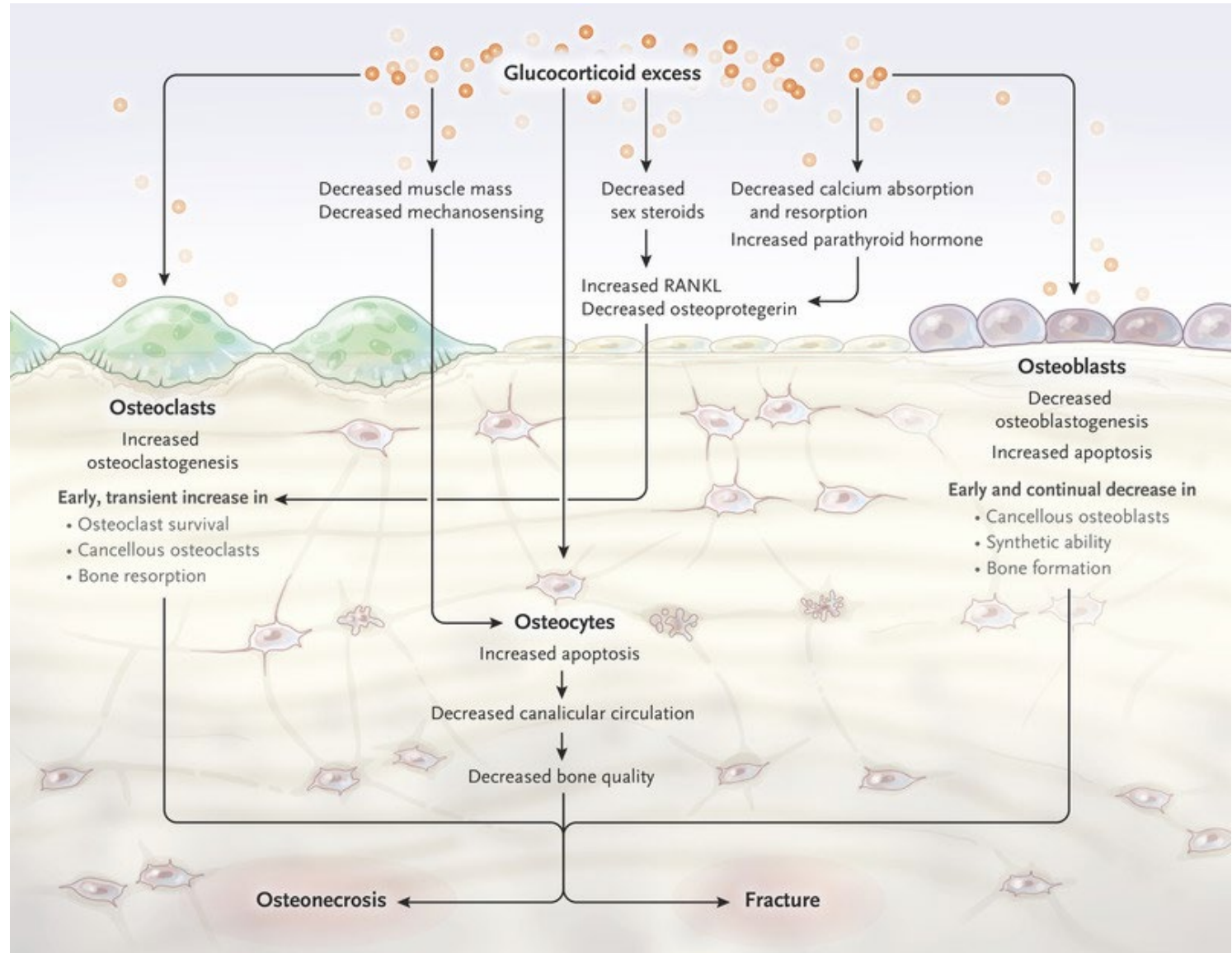


# Pathophysiology of age-related bone loss



- Increased rate of bone remodelling in both cancellous and cortical bone
- Negative remodelling balance

# Pathophysiology of glucocorticoid-induced osteoporosis



**Table 4. Common secondary causes of osteoporosis**

| Cause                            | Disorder/drug class                                                                                                                                                                                                               |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Endocrine disorders              | Hyperthyroidism<br>Hyperparathyroidism<br>Cushing syndrome<br>Hypogonadism<br>Hyperprolactinaemia<br>Anorexia nervosa                                                                                                             |
| Disorders of calcium metabolism  | Vitamin D deficiency<br>Hypercalciuria                                                                                                                                                                                            |
| Gastrointestinal disorders       | Primary biliary cirrhosis<br>Pancreatic diseases<br>Low acidity states: gastrectomy, gastric bypass, pernicious anaemia<br>Haemochromatosis<br>Malabsorption syndrome: inflammatory bowel disease, coeliac disease                |
| Medications                      | Corticosteroids<br>Anticonvulsants<br>Proton pump inhibitors<br>Aromatase inhibitors<br>Thiazolidinediones<br>Anticoagulants<br>Depo-medroxyprogesterone<br>Chemotherapy<br>Immunosuppressants<br>Excessive thyroxine replacement |
| Miscellaneous medical conditions | Multiple myeloma<br>Thalassaemias<br>Rheumatoid arthritis<br>Stroke (hemi-osteoporosis)                                                                                                                                           |

# Osteomalacia

- Equivalent to rickets in children
- Abnormal histology: unmineralized osteoid
- Most common cause: vitamin D deficiency with very low level of serum 25(OH)D
- Alkaline phosphatase (ALP) high in osteomalacia (from secondary hyperparathyroidism), normal in osteoporosis

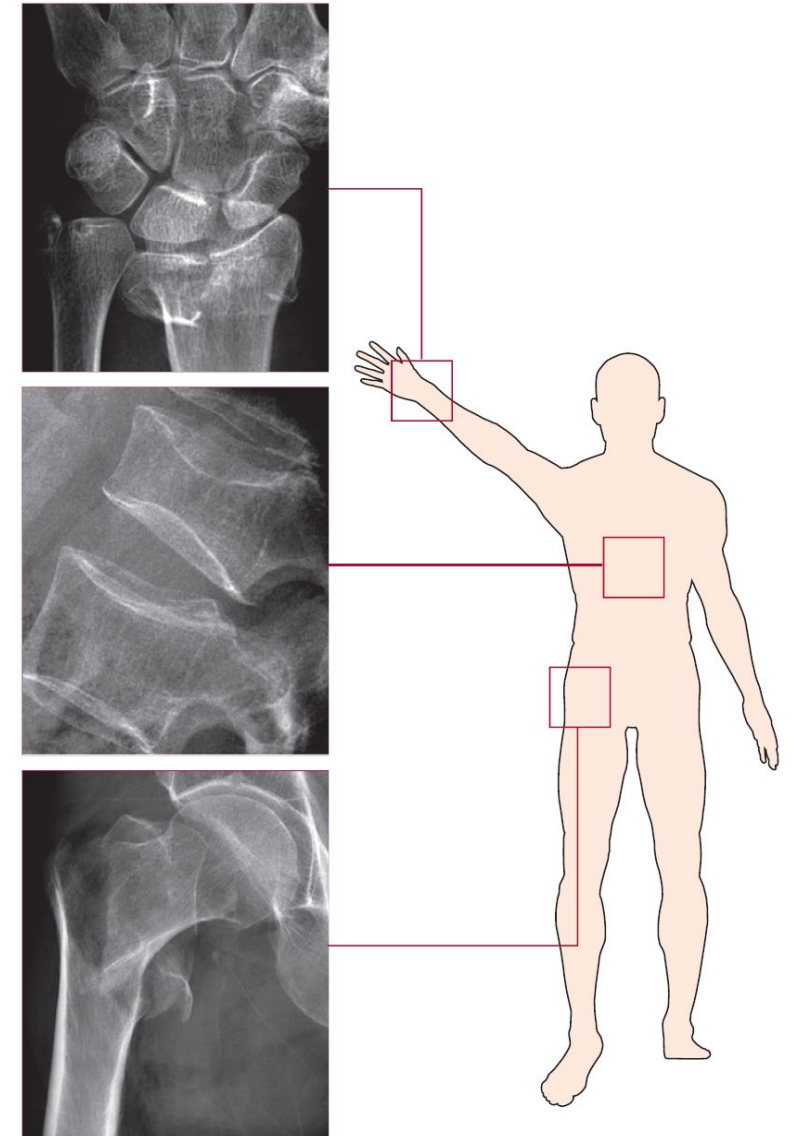
# Learning objectives

- Understand the common types of osteoporotic fractures and their clinical presentation
- Learn the epidemiology of osteoporotic fracture
- Understand the diagnosis of osteoporosis and measurement of bone density
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- Understand the prevention and treatment of osteoporosis
- Learn the management of osteoporotic fractures



# Clinical presentation

- Clinically silent until fractures occur
- Lifetime fracture risk could be as high as 40%
- Common sites of fractures
  - Wrist
  - Vertebra
  - Hip

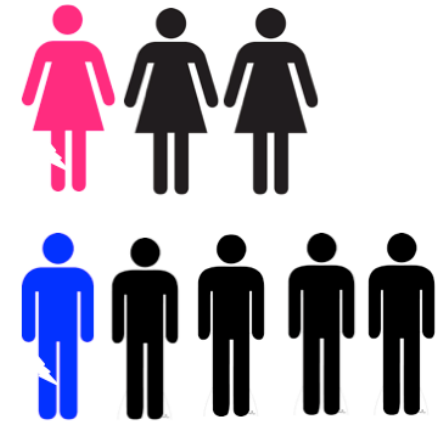


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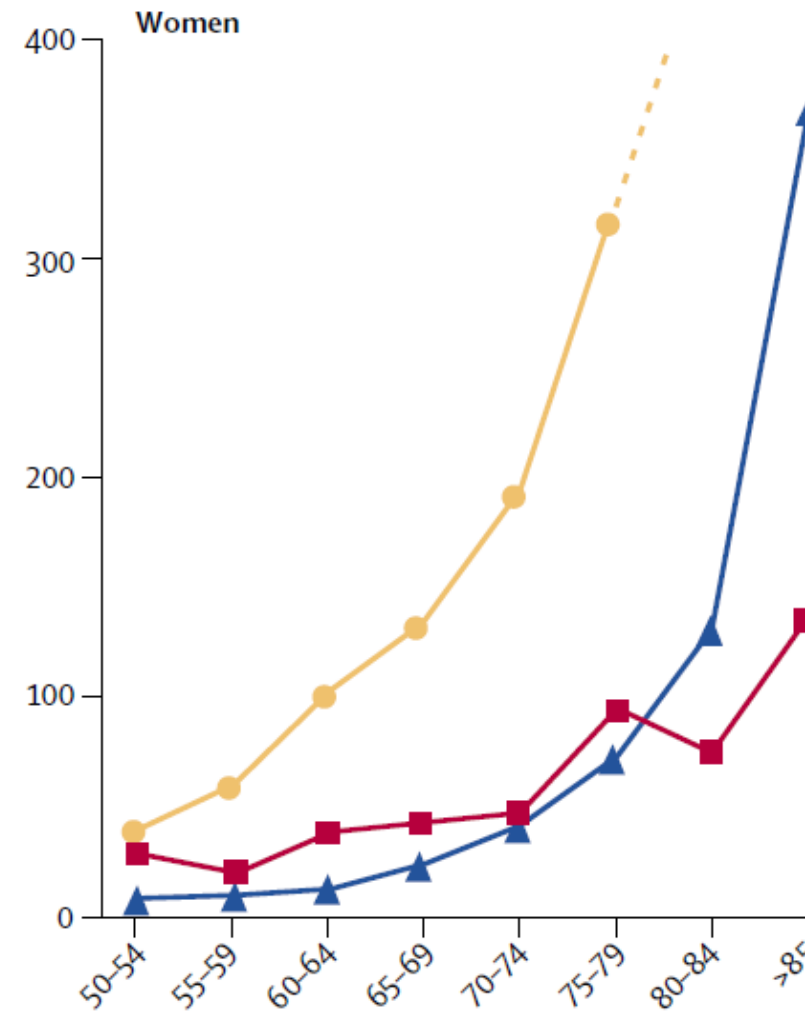
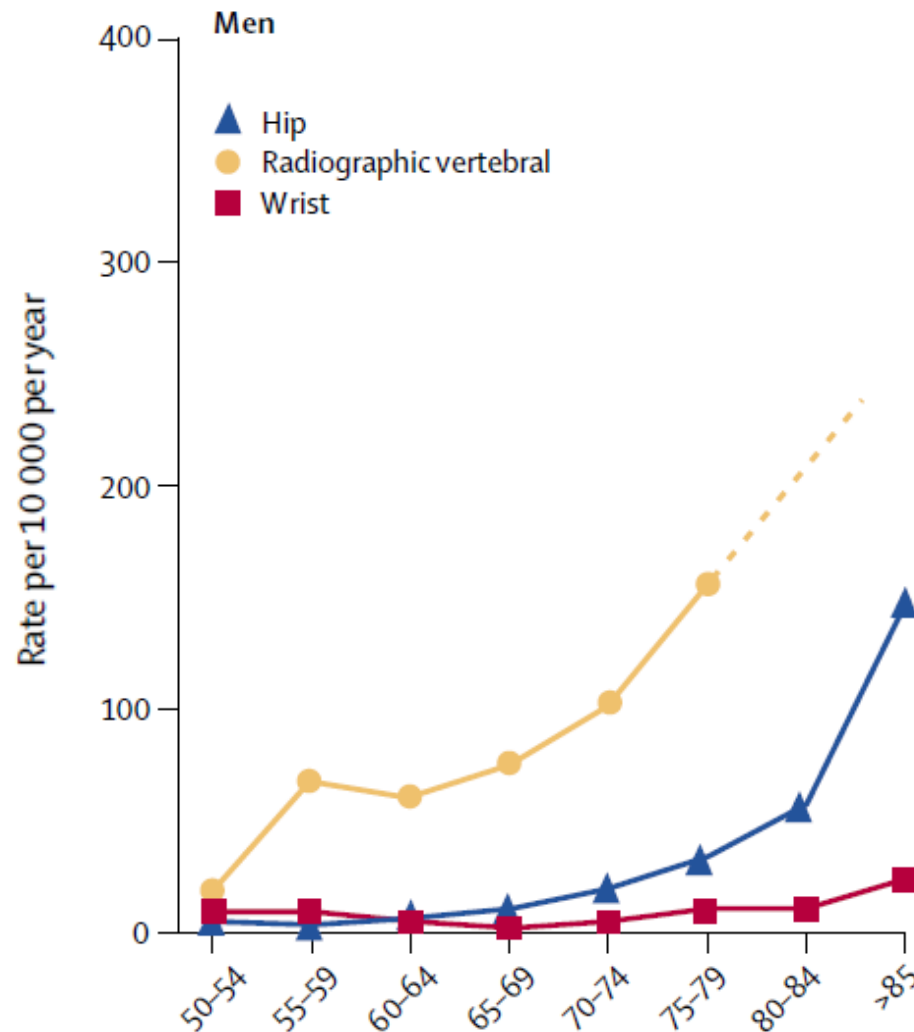
# Epidemiology

- The most common metabolic bone disease
- Osteoporotic fractures in lifetime
  - 1 in 3 postmenopausal women
  - 1 in 5 men aged  $\geq 50$  years
- 1 osteoporotic fracture occurs every 3 seconds
- Hip and vertebral fractures as the hallmark complications

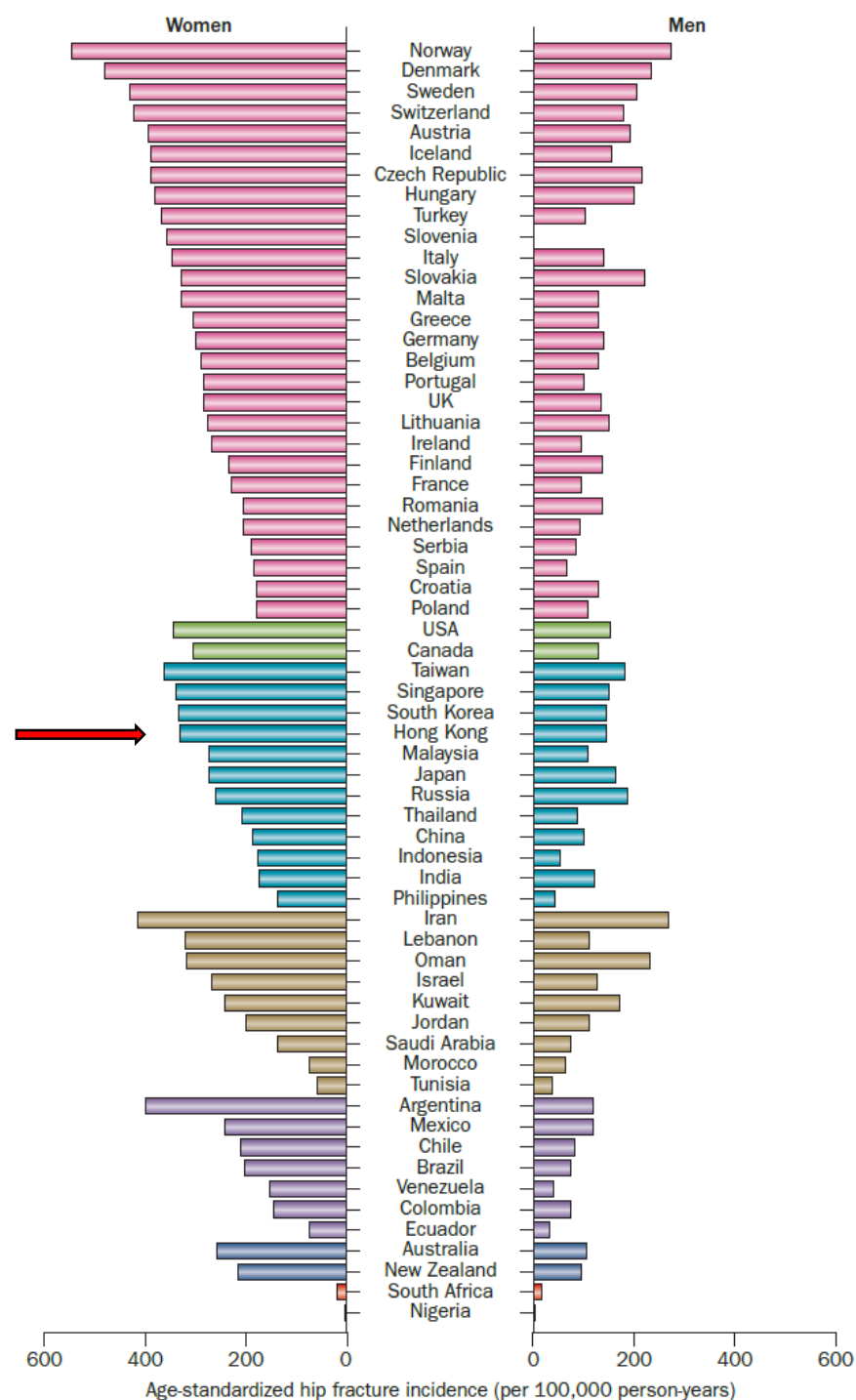


# Epidemiology

- Exponential increase as a function of age
- Female predominance



# Age-standardized hip fracture incidence



**MEN**

**149** per 100,000 person-years

**WOMEN**

**324** per 100,000 person-years

**TOTAL**

**230** per 100,000 person-years

**W:M RATIO**

**2.2**

Tsang SW, et al. *Osteoporos Int.* 2009;20(11):1939-1945.  
Cauley JA, et al. *Nat Rev Endocrinol.* 2014;10(6):338-351.



# Hip fractures: increased morbidity and mortality



One-year mortality rate: **20-30%**<sup>1</sup>



**40-60%** of survivors cannot regain pre-fracture level of mobility within one year<sup>2</sup>



**20-60%** of previously independent patients require assistance with at least one ADL<sup>2</sup>

1. Cheng SY, et al. *Osteoporos Int.* 2011;22(10):2575-2586.

2. Dyer SM, et al. *BMC Geriatr.* 2016;16(1):158.

# Vertebral fractures: increased morbidity and mortality



20% excess mortality 5-year post-fracture<sup>1</sup>



Acute and chronic back pain<sup>2</sup>



Deformity and height loss<sup>2</sup>



Diminished quality of life<sup>2</sup>

1. Cooper C, et al. *Am J Epidemiol.* 1993;137(9):1001-1005.

2. Kendler DL, et al. *Am J Med.* 2016;129(2):e1-10.

# Learning objectives

- Understand the common types of osteoporotic fractures and their clinical presentation
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# How is osteoporosis diagnosed?

## Operational definition

World Health Organization (1994): bone mineral density (BMD)-based diagnosis

| Diagnostic Category               | Definition                                                                                                                              |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Normal                            | BMD within 1 SD of the young adult mean ( <b>T-score <math>\geq -1.0</math></b> )                                                       |
| Osteopenia                        | BMD >1 SD below the young adult mean, but <2.5 SD below this value ( <b>T-score <math>&lt; -1.0</math> and <math>&gt; -2.5</math></b> ) |
| Osteoporosis                      | BMD $\geq 2.5$ SD below the young adult mean ( <b>T-score <math>\leq -2.5</math></b> )                                                  |
| Severe (established) osteoporosis | BMD $\geq 2.5$ SD below the young adult mean ( <b>T-score <math>\leq -2.5</math></b> ) in the presence of $\geq 1$ fragility fractures  |

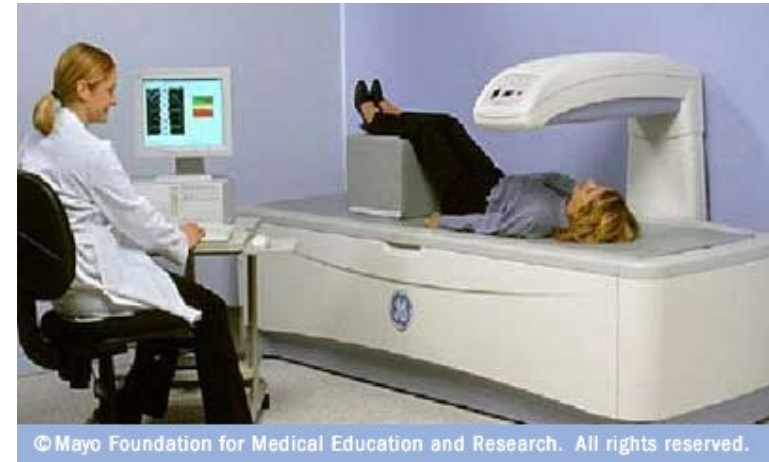
## Clinical definition

Low-trauma / low-energy fracture in a postmenopausal woman or elderly

- Defined as a fracture that occurs from a fall from standing height or lower

# Dual-energy x-ray absorptiometry (DXA)

- Gold standard: the only means for diagnostic classification according to WHO diagnostic criteria
- Central DXA at hip and spine are the recommended sites
  - BMD measurements at these regions of interest are the best predictor of fracture risk for the corresponding sites
- For every reduction of 1 SD, fracture risk increases by 1.5–2 times overall



**Relative risk of fracture for each age-adjusted standard deviation decrease in BMD**

| Skeletal site of BMD measurement | Hip fracture | Vertebral fracture |
|----------------------------------|--------------|--------------------|
| Distal radius                    | 1.8          | 1.7                |
| Proximal radius                  | 2.1          | 2.2                |
| Calcaneus                        | 2.0          | 2.4                |
| Spine                            | 1.6          | 2.3                |
| Femoral neck                     | 2.6          | 1.8                |

BMD: bone mineral density.

# BMD T-score

$$\text{T-score} = \frac{\text{Measured BMD} - \text{young adult mean BMD}}{\text{Young adult population SD}}$$

\* Gender- and ethnicity-matched population



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# Clinical assessment



# Clinical risk factors of osteoporosis

- Female sex
- Increasing age
- Low body weight (<45kg)
- Family history of osteoporosis or fragility fracture
- Premature menopause (age <40) or early menopause (age 40–45)
- Low calcium intake
- Lack of exercise or sedentary lifestyle
- Smoking
- Excessive alcohol intake (>3 standard drinks per day)
- Lack of sun exposure
- Prolonged immobilization

# Initial evaluation of osteoporosis

## Detailed history of clinical risk factors for osteoporosis and/or fractures

- Smoking and alcohol
- Personal / parental history of fracture(s)
- Fall risk
- Estimate daily dietary calcium intake
- Past health: prolonged sex hormone deficiency (age of menopause, use of HRT), type 1 diabetes, anorexia nervosa, rheumatological diseases, GI disease / malabsorption syndrome (gastrectomy, intestinal bypass, Crohn's disease, coeliac disease)
- Drug history
  - Increased risk of osteoporosis: glucocorticoid, PPI, anticonvulsants, thiazolidinediones
  - Increased risk of fall: hypnotics, diuretics

# Fracture Risk Assessment Tool (FRAX<sup>®</sup>)

- Launched in 2008
- Simple ethnic-specific web-based tool that integrates clinical information in a quantitative manner to predict a 10-year probability of major osteoporotic fracture and hip fracture for both women and men
- Applies only to postmenopausal women or men aged  $\geq 50$  years who have NOT been treated
- Does NOT apply to pre-menopausal women or younger adults
- <http://www.shef.ac.uk/FRAX/>

The screenshot shows the FRAX Fracture Risk Assessment Tool web interface. The header is red with the FRAX logo and title. A navigation bar includes links for Home, Calculation Tool, Paper Charts, FAQ, and References, along with a language dropdown set to English. The main section is titled 'Calculation Tool' and instructs users to answer questions to calculate the 10-year probability of fracture with BMD. The 'Questionnaire' section includes fields for Country (Hong Kong), Name/ID, Age (70), Sex (Female), Weight (60 kg), Height (165 cm), and various clinical factors like Previous Fracture, Parent Fractured Hip, Current Smoking, Glucocorticoids, and Rheumatoid arthritis. It also includes checkboxes for Secondary osteoporosis and Alcohol consumption. A 'Femoral neck BMD' section shows a T-Score of -2.5. A red box displays the results: BMI: 22.0, The ten year probability of fracture (%) with BMD, Major osteoporotic fracture at 11%, and Hip Fracture at 4.2%. On the right, there are 'Weight Conversion' and 'Height Conversion' sections, and a footer with the number 00060901 and a note about the tool's use since June 2011.

FRAX<sup>®</sup> Fracture Risk Assessment Tool

Home Calculation Tool Paper Charts FAQ References English

### Calculation Tool

Please answer the questions below to calculate the ten year probability of fracture with BMD.

Country: Hong Kong Name/ID:  About the risk factors

#### Questionnaire:

1. Age (between 40 and 90 years) or Date of Birth  
Age:  Y:  M:  D:

2. Sex ☐ Male ☒ Female

3. Weight (kg)

4. Height (cm)

5. Previous Fracture ☒ No ☐ Yes

6. Parent Fractured Hip ☒ No ☐ Yes

7. Current Smoking ☒ No ☐ Yes

8. Glucocorticoids ☒ No ☐ Yes

9. Rheumatoid arthritis ☒ No ☐ Yes

10. Secondary osteoporosis ☒ No ☐ Yes

11. Alcohol 3 or more units/day ☒ No ☐ Yes

12. Femoral neck BMD (g/cm<sup>2</sup>)  
T-Score

**BMI: 22.0**  
**The ten year probability of fracture (%) with BMD**

|                    |     |
|--------------------|-----|
| Major osteoporotic | 11  |
| Hip Fracture       | 4.2 |

If you have a TBS value, click here:

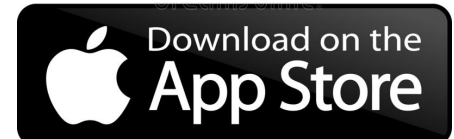
#### Weight Conversion

Pounds  kg

#### Height Conversion

Inches  cm

00060901  
Individuals with fracture risk assessed since 1st June 2011



# Physical examination

| System  | Findings                      | Potential clinical implications        |
|---------|-------------------------------|----------------------------------------|
| General | Loss of height / kyphosis     | Vertebral fracture                     |
|         | Weight gain / central obesity | Cushing's syndrome                     |
|         | Weight loss                   | Malabsorption                          |
|         | Goitre / eye signs            | Thyrotoxicosis                         |
|         | Joint disease                 | Rheumatoid arthritis                   |
|         | Eunuchoid                     | Hypogonadism                           |
| Teeth   | Poor dentition                | Increased risk of osteonecrosis of jaw |
| CNS     | Muscle weakness               | Vitamin D deficiency                   |
|         | Gait and balance              | High fall risk                         |



# Laboratory investigations

| Diagnostic test                                                                                                                                                                         | To detect or exclude                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| 1. Complete blood count                                                                                                                                                                 | Anaemia (myeloma / coeliac disease)   |
| 2. Renal and liver function tests                                                                                                                                                       | Renal or liver failure, alcohol abuse |
| 3. Calcium, phosphate and parathyroid hormone (PTH)                                                                                                                                     | Primary hyperparathyroidism, myeloma  |
| 4. Alkaline phosphatase (ALP)                                                                                                                                                           | Osteomalacia, Paget's disease         |
| 5. 25-hydroxyvitamin D level (25OHD)                                                                                                                                                    | Vitamin D deficiency, osteomalacia    |
| 6. TSH $\pm$ fT4                                                                                                                                                                        | Hyperthyroidism                       |
| 7. Testosterone (in men)                                                                                                                                                                | Male hypogonadism                     |
| 8. Fasting glucose and HbA1c                                                                                                                                                            | Diabetes mellitus                     |
| 9. 24hr urine for calcium                                                                                                                                                               | <i>Hypercalciuria</i>                 |
| <b><i>Selected patients</i></b>                                                                                                                                                         |                                       |
| <ul style="list-style-type: none"><li>Bone turnover markers, serum protein electrophoresis, serum and/or urinary cortisol, prolactin, anti-tissue transglutaminase antibodies</li></ul> |                                       |



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# Whom to treat?

Indications for therapy in postmenopausal women and men age  $\geq 50$  years

- Prior hip or vertebral fracture
- BMD T-score  $\leq -2.5$  at the spine or proximal femur
- BMD T-score between -1 and -2.5 (low bone mass) and one of followings:
  - 10-year risk of major fracture  $\geq 20\%^*$
  - 10-year risk of hip fracture  $\geq 3\%^*$

*\* calculated by FRAX<sup>®</sup> algorithm*



# Management of osteoporosis

The objective of treating osteoporosis is to reduce the likelihood of fragility fractures by strengthening the skeleton or decreasing fall frequency, or both

# Management of osteoporosis



Non-pharmacological



Pharmacological

# Non-pharmacological management

- Healthy balanced diet rich in calcium and vitamin D
- Regular weight-bearing and muscle-strengthening exercises
- Avoidance of smoking and excessive alcohol intake
- Adequate sunlight exposure

# Calcium intake

- 800–1200 mg/day, ideally achieved through dairy products
- Calcium supplements reasonable if dietary intake <800mg/day
  - Calcium carbonate
  - Calcium citrate
- Taken with food as gastric acidity promotes calcium absorption
- Common adverse effects are dyspepsia and constipation
  - May settle with increasing fluid intake or dietary fibre consumption
  - May try another type or brand of calcium supplement

**Table 6. Calcium content of some common local foods**

| Food item       | Calcium content (mg per 100 g) |
|-----------------|--------------------------------|
| Cheese          | 675                            |
| Sardine         | 400                            |
| Bean curd sheet | 330                            |
| Almond          | 250                            |
| Yogurt          | 170                            |
| Tofu            | 150                            |
| Milk            | 120                            |
| Broccoli        | 75                             |

# Vitamin D

- Important roles
  - Promotes calcium absorption in gut
  - Maintains adequate serum calcium and phosphate levels for normal bone mineralization
  - Important in neuromuscular function
- Major source of vitamin D: synthesis of cholecalciferol (vitamin D3) from its precursors in the skin under the effect of UV light
  - Around 10-15 minutes of sunlight exposure every day to the exposed areas over the face, hands, and arms is considered adequate to meet daily requirement

# Vitamin D

- Mean vitamin D requirement: 800-1000 IU/day
- Some may need higher dose (up to 2000 IU/day)
  - Obese
  - Limited sun exposure
  - Malabsorption
- Major circulating form of vitamin D: 25-hydroxyvitamin D = 25(OH)D
  - Summation of oral intake and sun exposure
  - To classify patients into degree of vitamin D sufficiency
- Desired target: 25(OH)D level of 75 nmol/L (30 ng/mL)



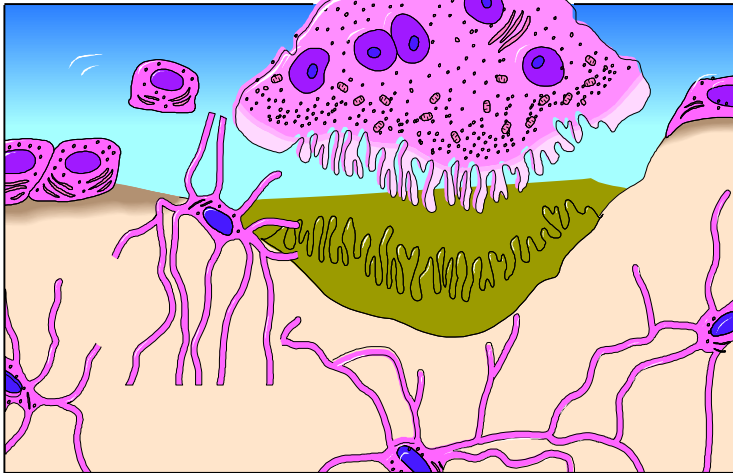
# Exercise

- Weight bearing exercises
  - Brisk walking
  - Jogging
  - Tai Chi
  - Dancing
  - Stair-climbing
- Retards bone loss
- Increases muscle mass / strength
- Improves balance and reduces fall
- Improves general well-being and quality of life



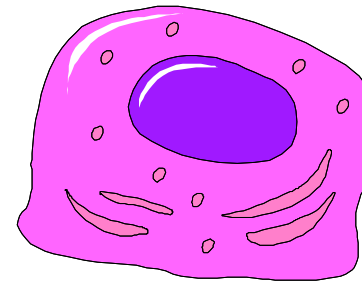
# Targets for pharmacological intervention

Osteoclast



**Inhibition of  
bone resorption**

Osteoblast



**Stimulation of  
bone formation**

# Anti-osteoporosis drugs

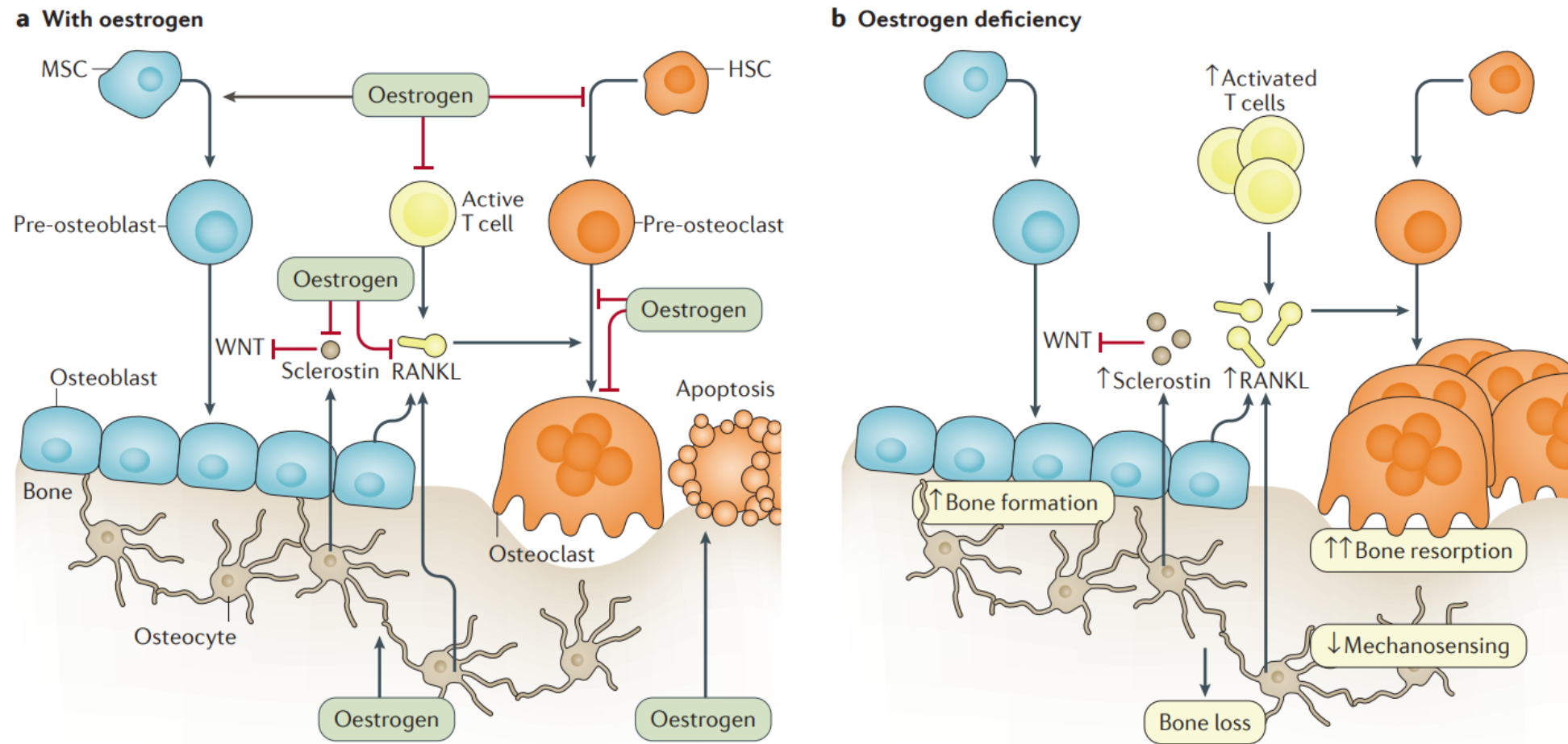
| Anti-resorptives             | Anabolic agents |
|------------------------------|-----------------|
| Hormonal replacement therapy | Teriparatide    |
| Raloxifene                   | Abaloparatide   |
| Bisphosphonates              | Romosozumab     |
| Denosumab                    |                 |

# Efficacy of anti-resorptives and anabolic agents

|                         | 2-year increase<br>Spine BMD | 2-year increase<br>Total hip BMD | RRR of vertebral<br>fracture | RRR of non-<br>vertebral fracture |
|-------------------------|------------------------------|----------------------------------|------------------------------|-----------------------------------|
| <b>Anti-resorptives</b> |                              |                                  |                              |                                   |
| PO raloxifene           | 2-3%                         | 1%                               | 50%                          | --                                |
| PO bisphosphonates      | 3-5%                         | 2-3%                             | 40-53%                       | 0-20%                             |
| IV zoledronic acid      | 5-6%                         | 3-4%                             | 70%                          | 25%                               |
| SC denosumab            | 6-8%                         | 3-4%                             | 68%                          | 20%                               |
| <b>Anabolic agents</b>  |                              |                                  |                              |                                   |
| SC teriparatide         | 8-10%                        | 1.5-2%                           | 65-70%                       | 35%                               |
| SC abaloparatide        | 10%                          | 2-3%                             | 70-80%                       | 40%                               |
| SC romosozumab          | 11% (1 year)                 | 4% (1 year)                      | 48% (vs alendronate)         | 20% (vs alendronate)              |

Black DM, et al. *Lancet*. 1996;348(9041):1535-1541.  
 Neer RM, et al. *N Engl J Med*. 2001;344(19):1434-1441.  
 Black DM, et al. *N Engl J Med*. 2007;356(18):1809-1822.

Cummings SR, et al. *N Engl J Med*. 2009;361(8):756-765.  
 Miller PD, et al. *JAMA*. 2016;316(7):722-733.  
 Saag KG, et al. *N Engl J Med*. 2017;377(15):1417-1427.



**Figure 5 | Role of oestrogens in bone remodelling. a** | Physiological bone remodelling. Oestrogens enhance bone formation by stimulating osteoblast differentiation and function and suppress osteoclast differentiation and activity through direct cellular effects and reduced production of receptor activator of nuclear factor- $\kappa$ B ligand (RANKL) by activated T cells, mesenchymal stem cells (MSCs), osteoblasts and osteocytes. In addition, oestrogens are required for adequate mechanosensing and mechanotransduction by the osteocytes. **b** | Oestrogen deficiency. RANKL production is upregulated translating into a higher osteoclast number and activity. Concurrently, the function of individual osteoblasts is suppressed, partly by upregulation of the WNT inhibitor sclerostin, whereas the total number of osteoblasts increases. The role of osteocytes in mechanosensing is also impaired. HSC, haematopoietic stem cell.

# Hormonal replacement therapy (HRT)

- Inhibits bone resorption
  - ↓calcium excretion in urine
  - ↓cytokines production by stromal cells
- Additional advantages
  - CVD risk factors e.g. cholesterol, but outcome events (i.e. MI, stroke) not reduced
  - ↓menopausal symptoms
- Disadvantages
  - Endometrial cancer (additional progestogens if uterus intact)
  - Slight ↑risk of breast cancer
  - Venous thromboembolism
- Mainly an option for young postmenopausal women especially for those with climacteric symptoms

# Raloxifene



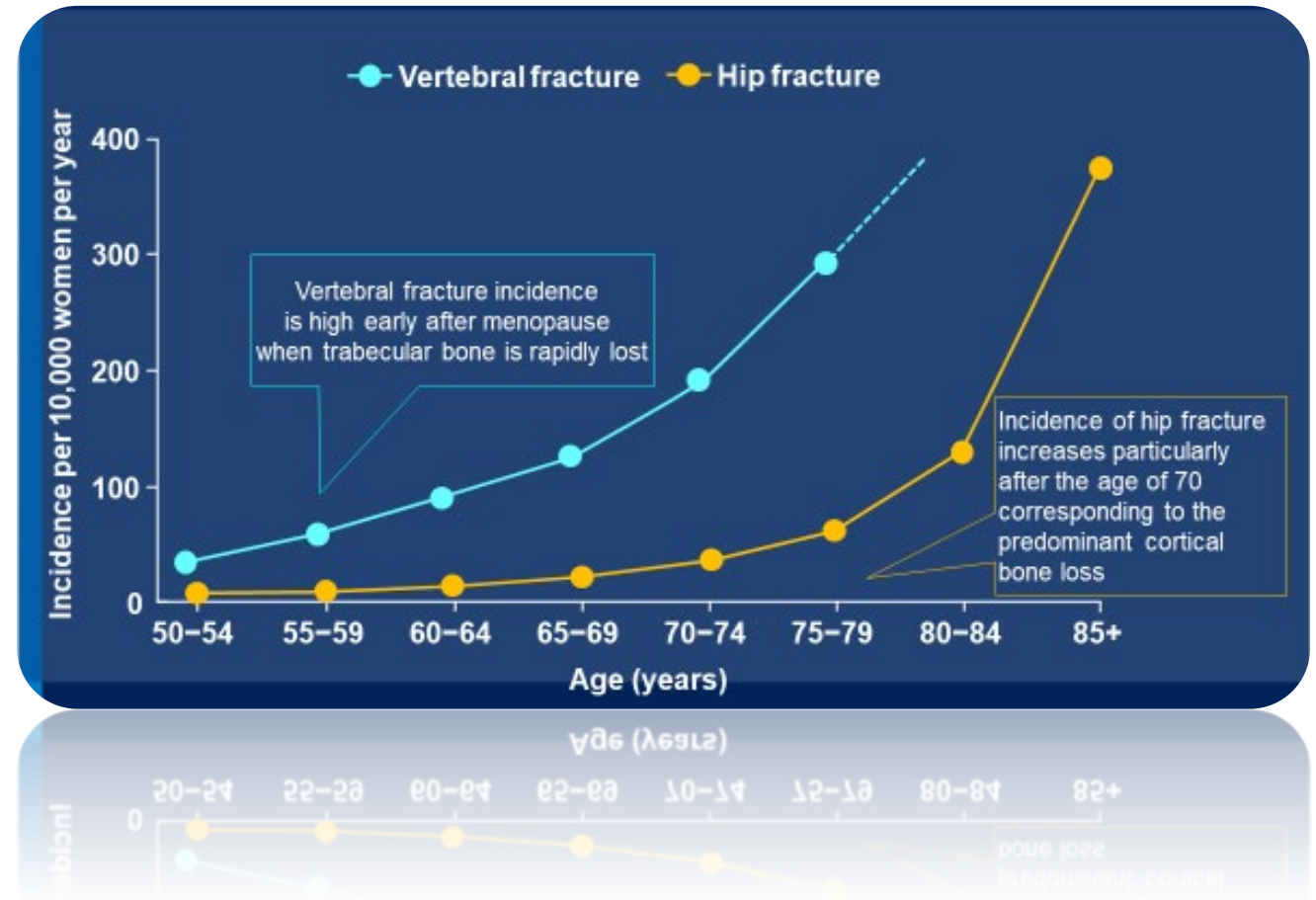
- Selective estrogen receptor modulators (SERMs)
- **Oestrogen-agonistic effects on bone**
  - 40% ↓ vertebral fracture risk, but not for non-vertebral fractures
- **Oestrogen-antagonistic effects on endometrium and breast**
  - ↓ incidence of invasive ER+ breast cancer
- Side effects: venous thromboembolism, hot flushes, leg cramps
- Should be discontinued in the event of an illness or a condition leading to a prolonged period of immobilization

Delmas PD, et al. *J Clin Endocrinol Metab.* 2002 Aug;87(8):3609-17.

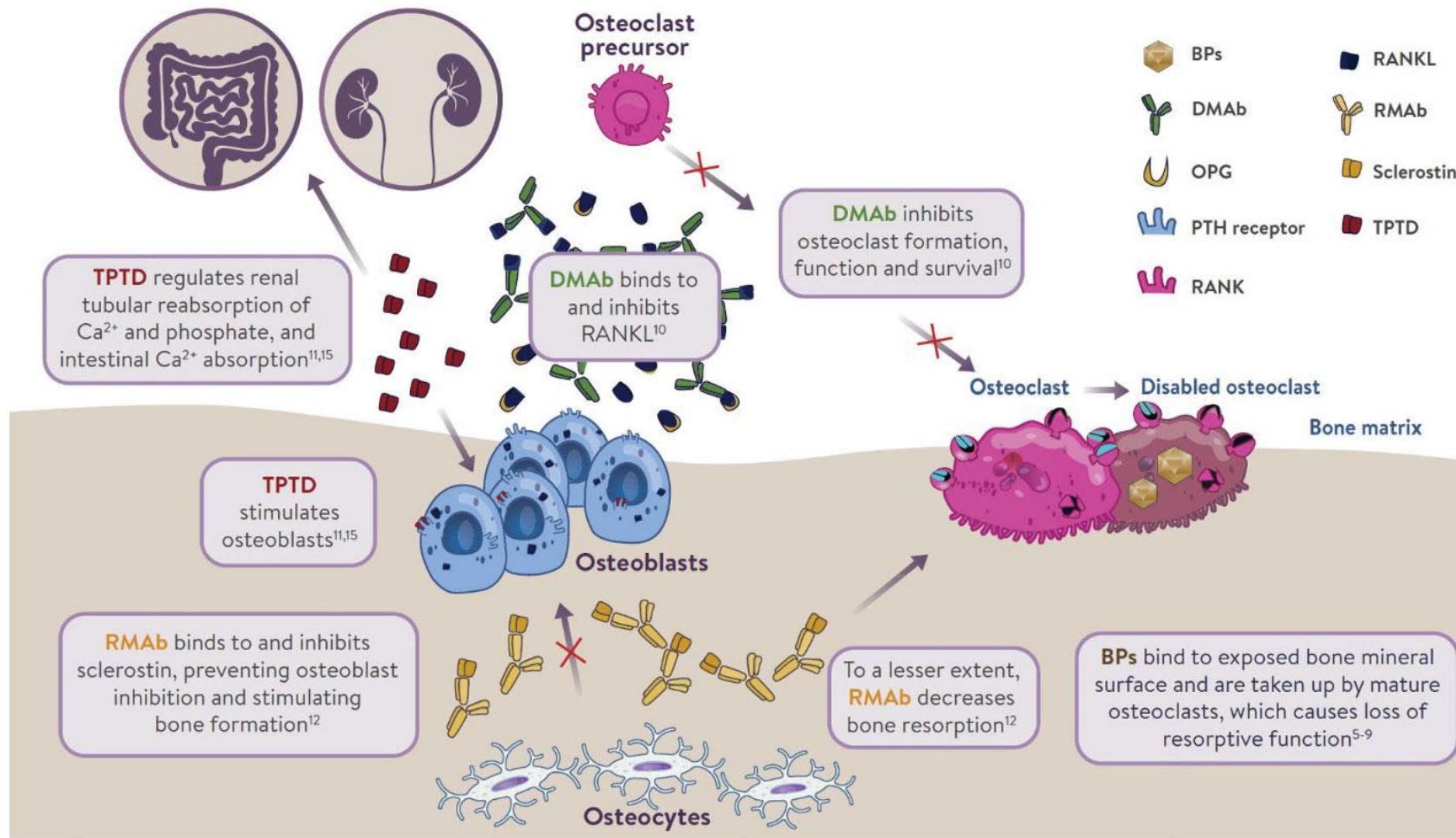
Eastell R, et al. *J Clin Endocrinol Metab.* 2019;104(5):1595-1622.

# Raloxifene

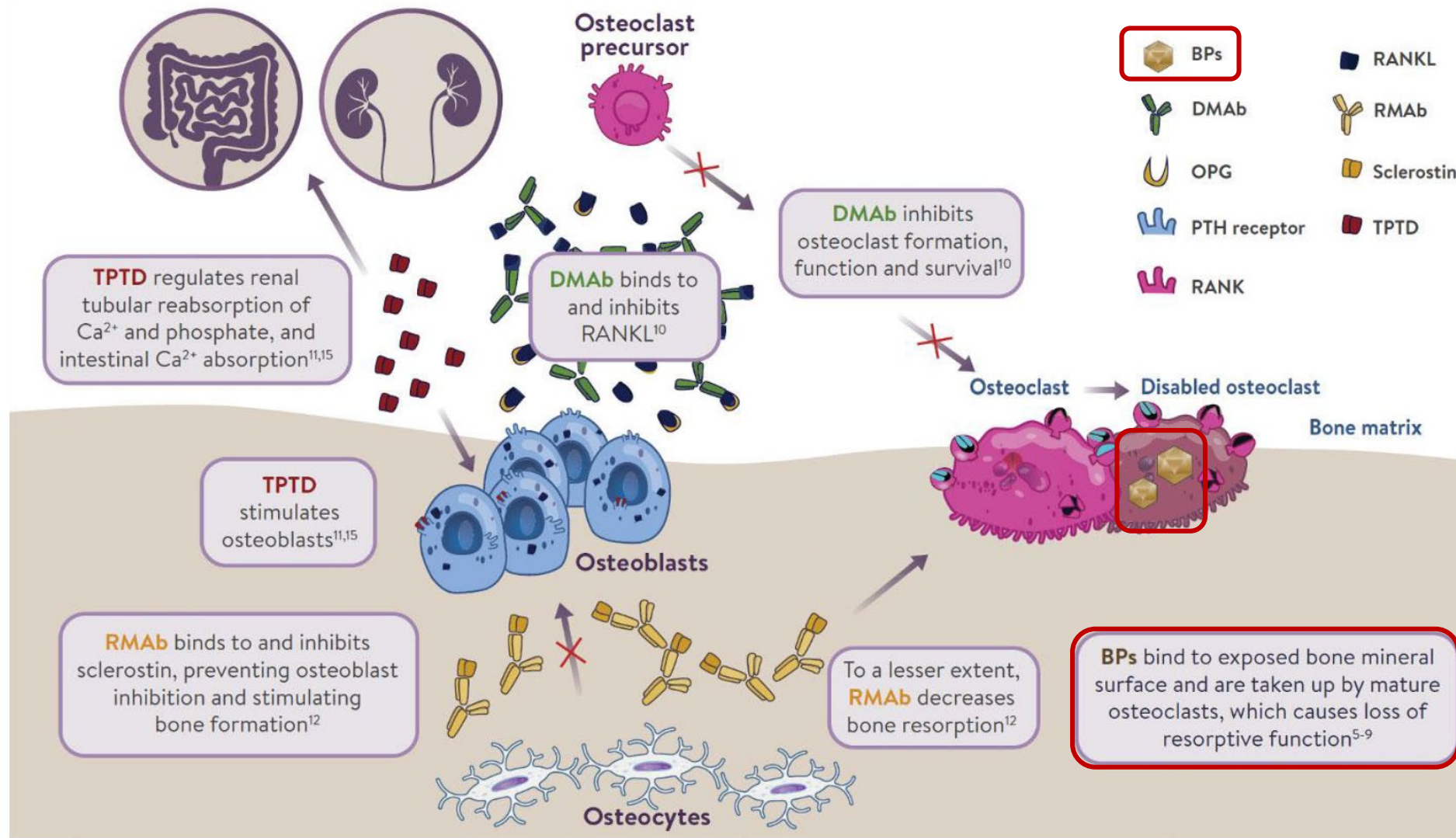
- Preferable in young post-menopausal women when the risk of hip fracture is not particularly high
- As patients age, may need to switch to more potent agents as the risk of hip fracture becomes higher







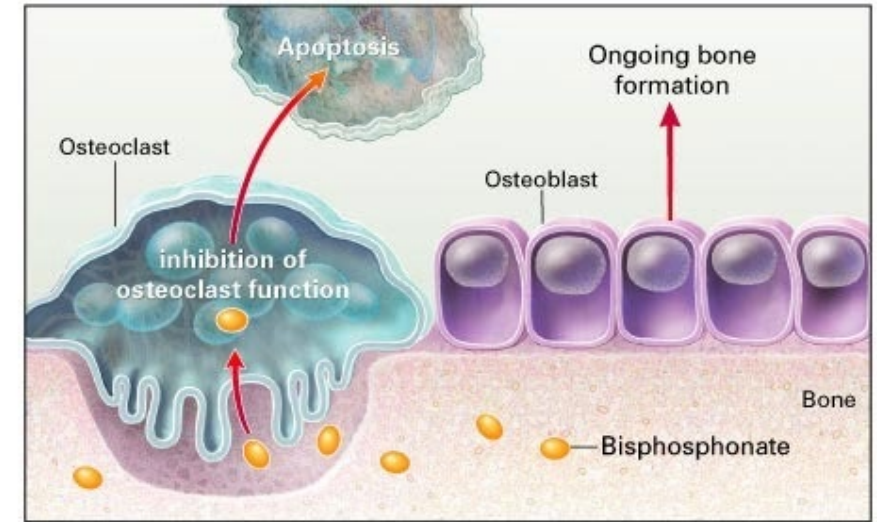
**Fig. 1.** Mechanism of action of osteoporosis therapies. BP, bisphosphonate; DMAB, denosumab; OPG, osteoprotegerin; PTH, parathyroid hormone; RANK, receptor activator of nuclear factor kappa-B; RANKL, receptor activator of nuclear factor kappa-B ligand; RMAb, romosozumab; TPTD, teriparatide.



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# Bisphosphonates

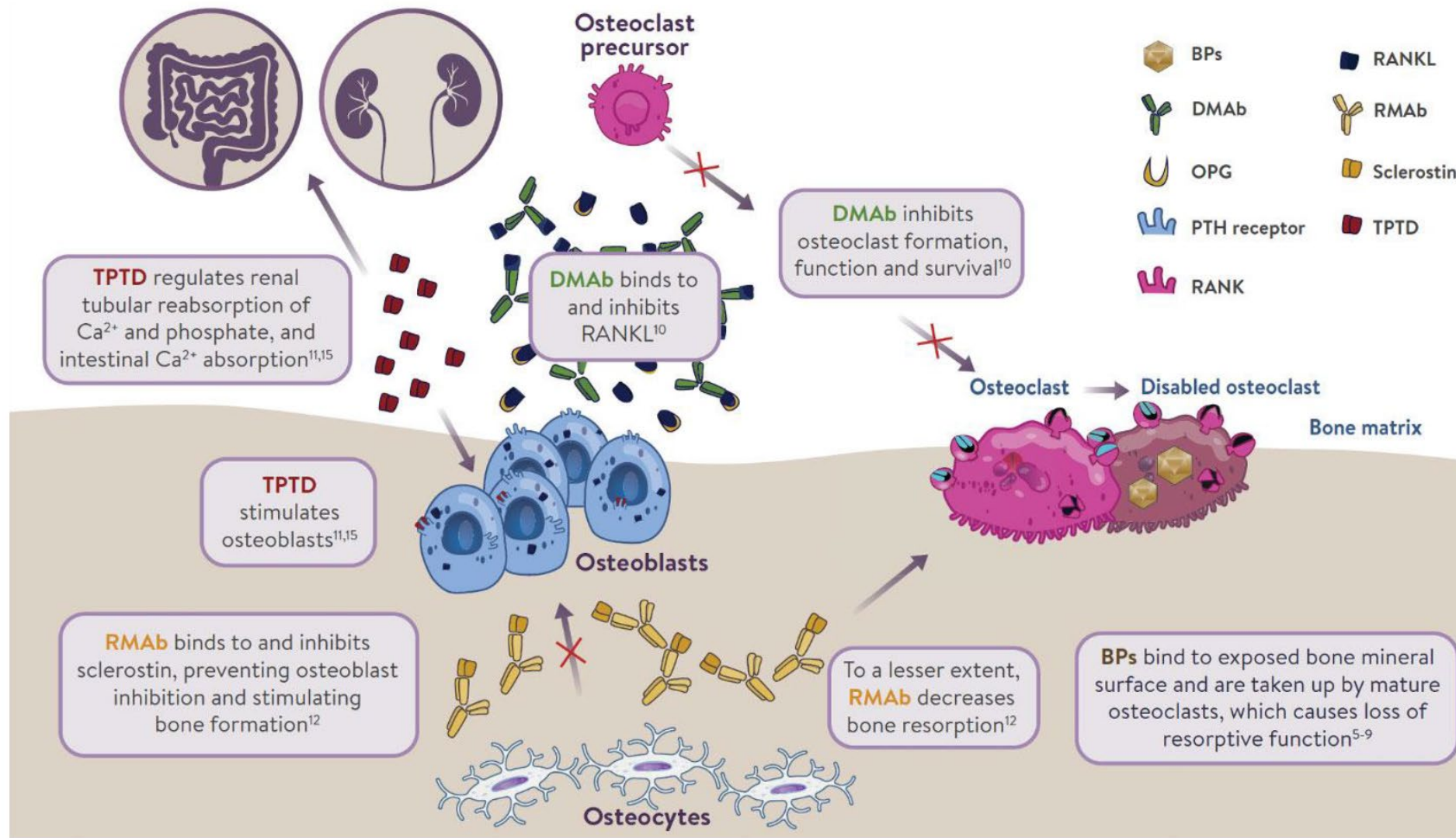
- Mainstay of pharmacotherapy
- Potent anti-resorptives
- Derivatives of pyrophosphates
- Inhibit osteoclast activity, specific inhibition of bone resorption
- Bind onto surface of osteoclast, inhibit farnesyl pyrophosphate synthase (FPPS), the key enzyme in the mevalonate pathway and induce apoptosis of osteoclast
- Renal excretion: contraindicated when  $\text{eGFR} < 30 \text{ mL/min}$
- Poor intestinal absorption
- Oral and IV preparations available
  - Oral: alendronate, risedronate, ibandronate
  - IV: zoledronic acid



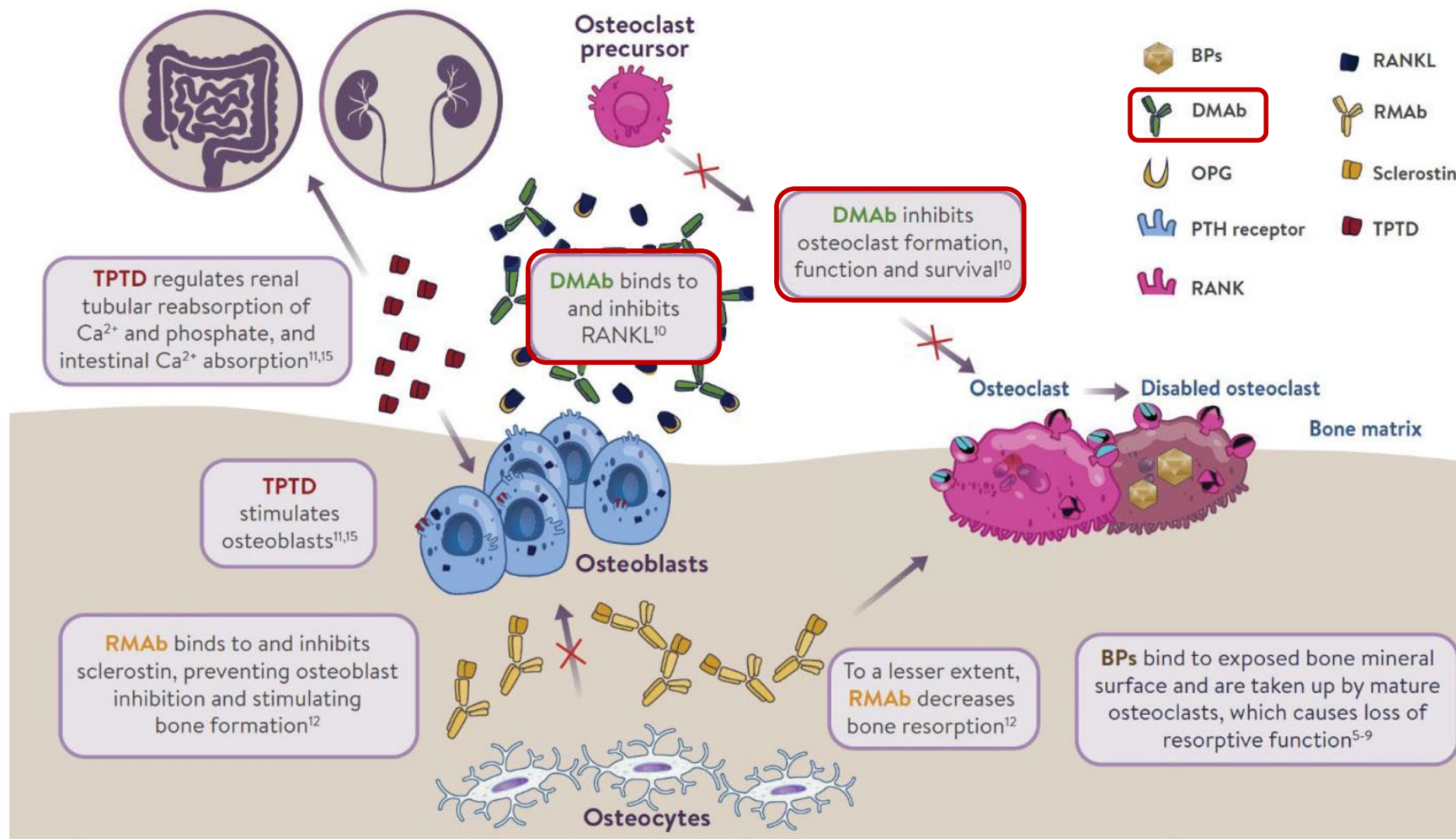


# Bisphosphonates

- Side effects
  - Upper gastrointestinal irritation (for oral bisphosphonates)
  - Acute phase reaction (for IV bisphosphonates esp. 1<sup>st</sup> dose)
  - Disabling muscle / bone pain – occasional
- Serious side effects – rare
  - Osteonecrosis of the jaw (ONJ): 1–10 in 100,000
  - Atypical femur fracture (AFF): 20–50 in 100,000



**Fig. 1.** Mechanism of action of osteoporosis therapies. BP, bisphosphonate; DMAB, denosumab; OPG, osteoprotegerin; PTH, parathyroid hormone; RANK, receptor activator of nuclear factor kappa-B; RANKL, receptor activator of nuclear factor kappa-B ligand; RMAb, romosozumab; TPTD, teriparatide.

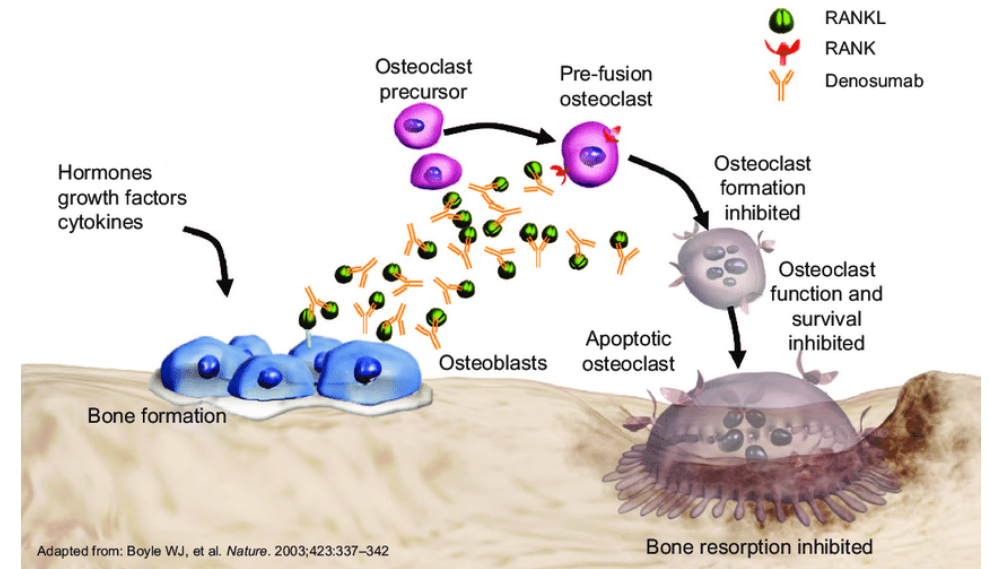


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# Denosumab

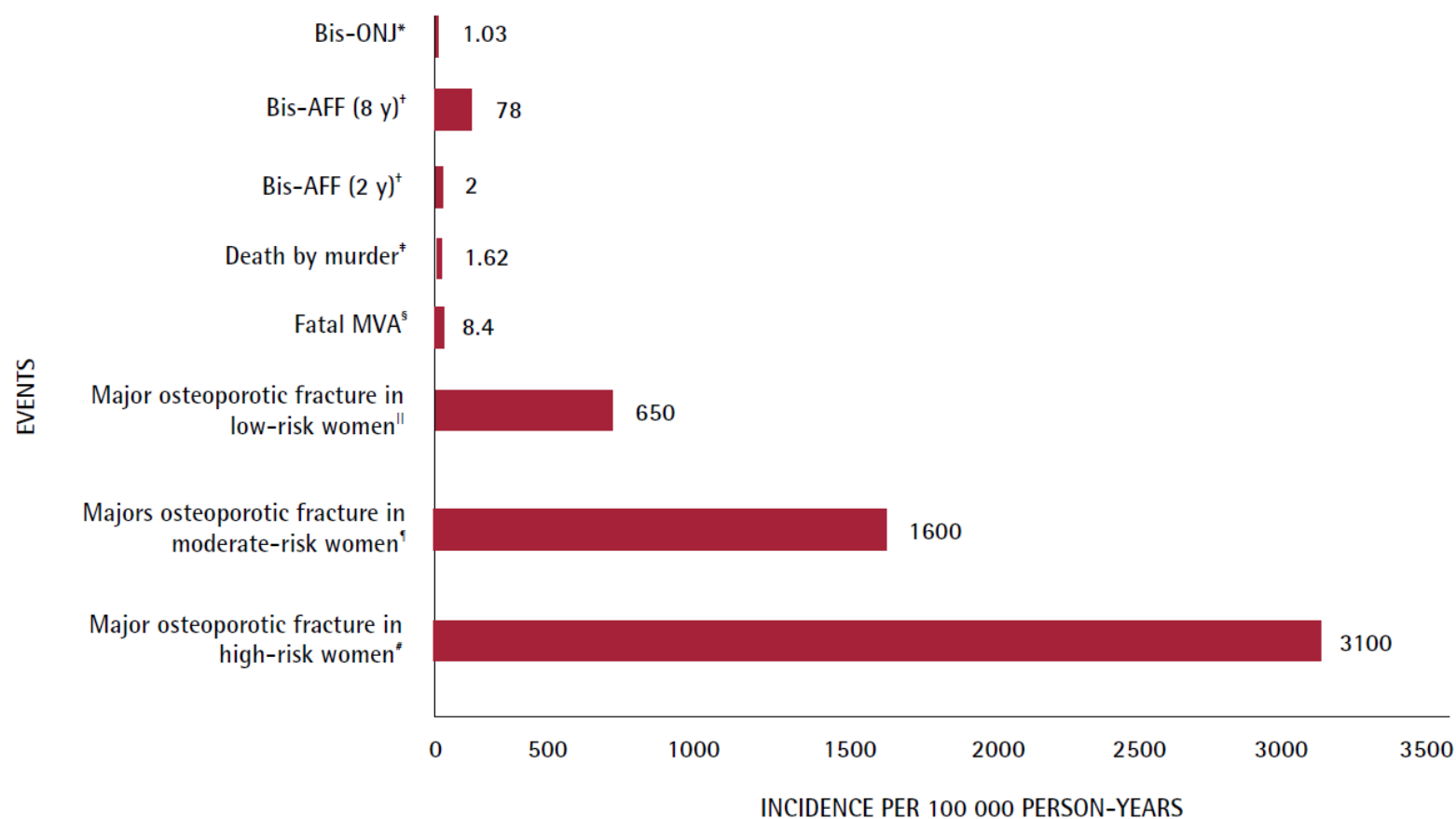
- Human monoclonal antibodies to RANKL
- Very potent antiresorptive drug
- Interfere with RANKL and decrease osteoclast differentiation and activation
- Given as SC injection every 6 months
- Can be used in patients with renal impairment down to stage 4 CKD (eGFR >15mL/min)
- Side effects: rash
- Rare side effects: hypocalcaemia, ONJ, AFF



# Osteonecrosis of Jaw (ONJ) & Atypical Femoral Fractures (AFF)

## Exceedingly rare!

Figure 1. Risks of major osteoporotic fracture and other rare events





# Atypical femoral fractures (AFFs)

- Occur in the subtrochanteric region or diaphysis of the femur and have unique radiological features
  - Predominantly transverse fracture line
  - Periosteal callus formation
  - Minimal comminution
- Appear to be more common in patients exposed to long-term bisphosphonate therapy, with a higher risk (113 per 100 000 person-years) with >7-8 years of therapy
- Also reported to be associated with denosumab, with an even lower incidence

1 Bilateral atypical femoral fractures in an older woman after bisphosphonate therapy for 9 years\*



\* Note the characteristic findings of a predominantly transverse fracture line, periosteal callus formation and minimal comminution on the left, and the periosteal reaction on the lateral cortex on the right femur, indicating an early stress fracture. ♦

# Medication-related ONJ (MRONJ)

- Criteria
  - Exposed bone in the maxillofacial region that does not heal within 8 weeks after identification by a health care provider
  - Current or previous treatment with antiresorptive or antiangiogenic agents
  - No history of radiation therapy to the jaws or obvious metastatic disease to the jaws
- Incidence: 1:10000 to 1:100000
- Risk factors
  - Age >65 years
  - Periodontitis
  - Use of bisphosphonates >2 years
  - Smoking
  - Denture wearing
  - Diabetes mellitus
  - Invasive dental procedures e.g. dental extractions

Hellstein JW, et al. *J Am Dent Assoc.* 2011;142(11):1243-1251.

Ruggiero SL, et al. *J Oral Maxillofac Surg.* 2022;80(5):920-943.

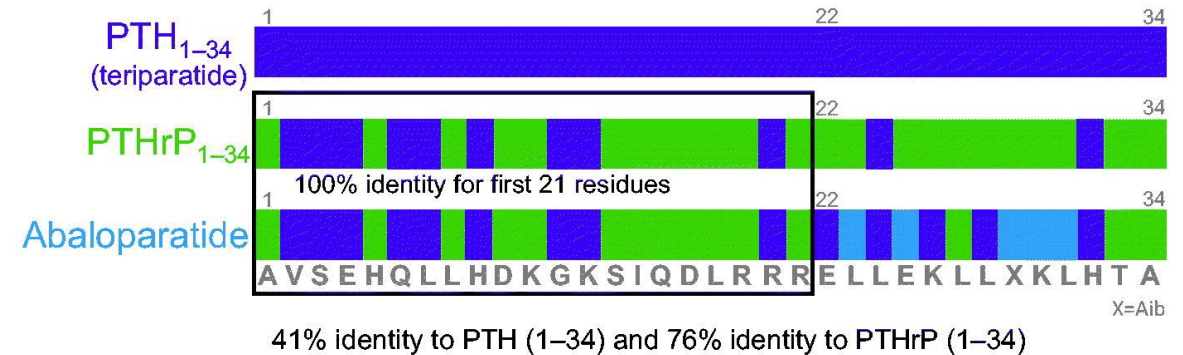
# Teriparatide



- 1-34 amino-terminal fragment of PTH
- High level, continuous PTH: bone resorption cortical > trabecular bone
- Low dose, intermittent PTH: anabolic action
- SC injection 20 microgram daily; limited to 2 years of treatment (concern with osteosarcoma in rats)
- Side effects: mild leg cramps, pain and bruising at injection site, hypercalcaemia
- Contraindications
  - History of skeletal malignancies/metastases or radiation therapy involving skeleton
  - Hyperparathyroidism

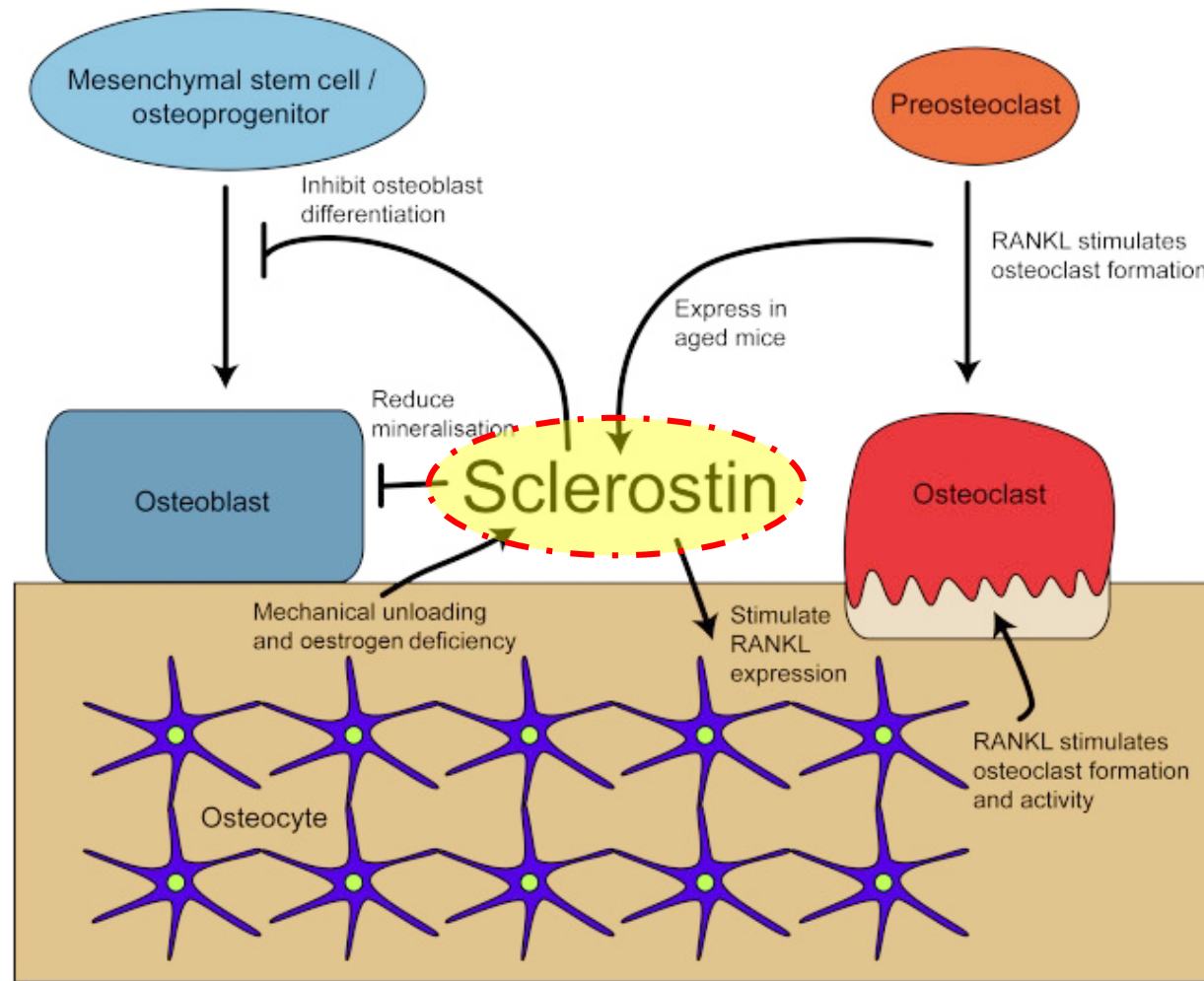
# Abaloparatide

- Synthetic analogue of PTH-related protein
- Similar skeletal effects as teriparatide
- Similar to teriparatide
  - Daily SC injection
  - Use limited to 2 years
- Hypercalcaemia less common in abaloparatide-treated than teriparatide-treated women (3.4% vs 6.4%)
- FDA-approved in April 2017
- Not available in Hong Kong

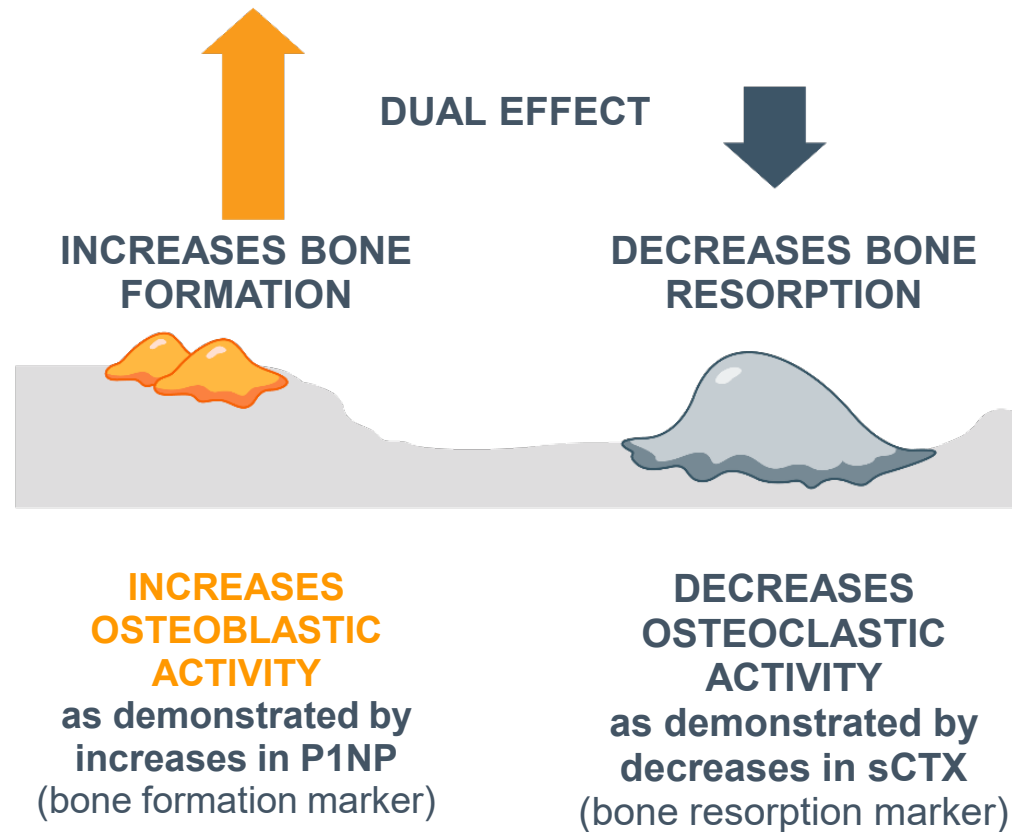


Compston JE, et al. *Lancet*. 2019;393(10169):364-376.  
Miller PD, et al. *Curr Med Res Opin*. 2020;36(11):1861-1872.

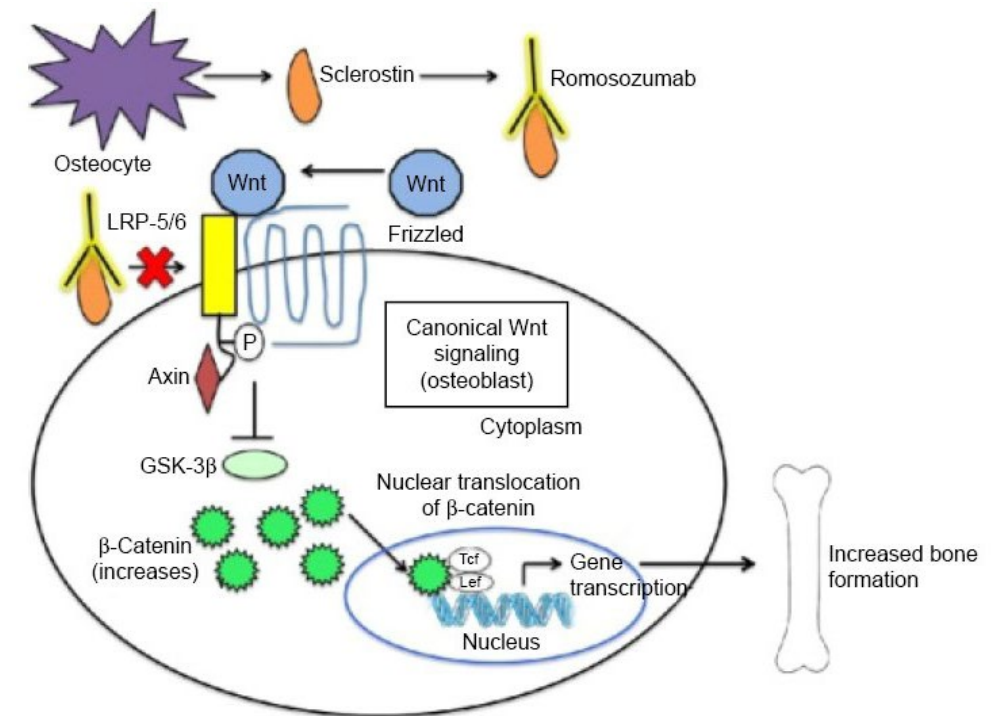
# The role of sclerostin in bone metabolism



# Romosozumab



Romosozumab is a humanised monoclonal antibody that binds and inhibits sclerostin, a regulatory factor in bone metabolism



# Romosozumab

- Humanized monoclonal antibody against sclerostin
- FDA approved in April 2019
- SC injection every month, for a duration of 1 year
- Side effects
  - Injection site reactions
  - Small risk of hypocalcaemia
  - ONJ, AFF (<0.1%)
- Caution in patients with high cardiovascular risks
- Contraindications: history of myocardial infarction or stroke

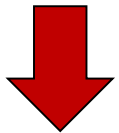




# Fracture Risk Stratification

## Very High Risk

- Multiple fractures
- Recent MOF  $\leq 24$  mo
- T-score  $< -3.0$
- Fracture on antiresorptive therapy



Anabolic agents

## High Risk

- Age  $\geq 65$  with T-score  $< -2.5$
- FRAX 10-y probability MOF  $\geq 20\%$  or HF  $\geq 3\%$
- Prior fracture  $> 24$  mo



Potent anti-resorptive  
(BP / Dmab)

## Low Risk

Age  $< 65$  with

- T-score  $< -2.5$
- No prior fracture

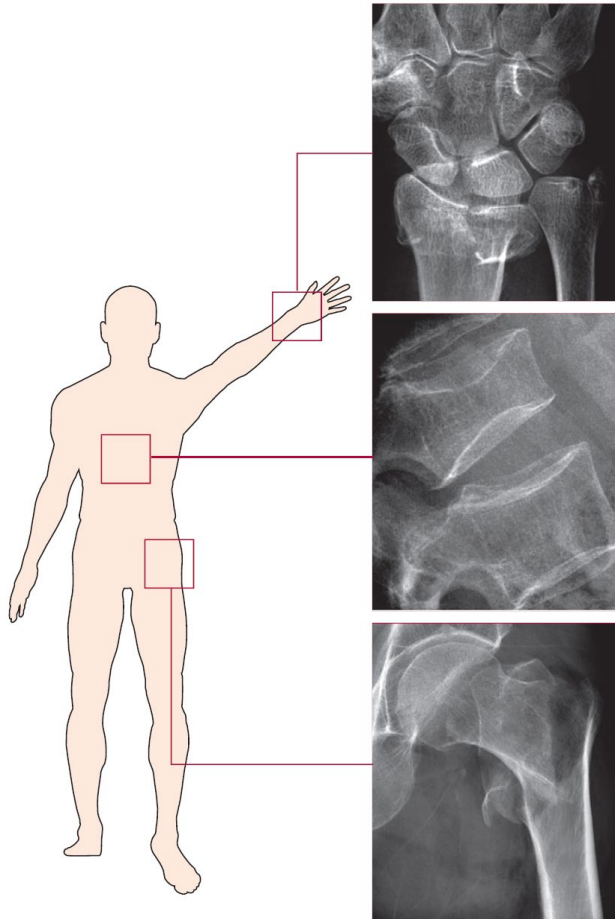


Raloxifene (for female)

# Learning objectives

- Understand the common types of osteoporotic fractures and their clinical presentation
- Learn the epidemiology of osteoporotic fracture
- Understand the diagnosis of osteoporosis and measurement of bone density
- Recognize the risk factors for osteoporosis
- Understand the prevention and treatment of osteoporosis
- Learn the management of osteoporotic fractures

# Management of osteoporotic fractures



Usually non-operative: manipulation and cast application for 4-6 weeks  
Operative management in selected patients

Conservative management: symptoms usually improve in 6 – 8 weeks  
Vertebral body augmentation procedures

Open surgical intervention

- Significant neurological deficits
- Progressive and severe kyphotic deformity
- Intractable pain

Operative management

Geriatric hip fracture clinical pathway

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# MEDICAL

Anticipating publication of  
**2024 OSHK Guideline**  
in 2024 Q2



OSTEOPOROSIS SOCIETY OF HONG KONG  
香港骨質疏鬆學會

2013 OSHK Guideline for  
Clinical Management of  
Postmenopausal Osteoporosis in Hong Kong

OSHK Task Group for Formulation of 2013 OSHK Guideline for Clinical Management of Postmenopausal Osteoporosis in Hong Kong, Ip TP, Cheung SK, et al. The Osteoporosis Society of Hong Kong (OSHK): 2013 OSHK guideline for clinical management of postmenopausal osteoporosis in Hong Kong. *Hong Kong Med J.* 2013;19 Suppl 2:1-40.

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